Mechanism/Pathophysiology Questions

Category: Mechanism/Pathophysiology **Total Questions:** 206 **Purpose:** Medical professional review of categorization accuracy

Review Instructions

For each question below, please mark:

- ✓ **Correctly categorized -** Question clearly belongs in this category
- ✗ **Miscategorized -** Question belongs in a different category (note which one)
- ? **Ambiguous -** Question could reasonably fit multiple categories

You don't need to review all questions! Even 20-30 per category is very helpful.

What 'Mechanism/Pathophysiology' Means

Questions asking about how drugs work, disease mechanisms, or biological pathways - the 'why' and 'how' questions.

Questions for Review

Question 1

Clinical Scenario:

A 47-year-old man with a history of diabetes mellitus presents for a primary care visit. His diabetes is well controlled on metformin, with fasting glucose concentrations between 110–150 mg/dl. His blood pressure on multiple office visits are between 115-130/75-85 mmHg. Today his temperature is 98°F (36.7 °C), blood pressure is 125/80 mmHg, pulse is 86/min, and respirations are 15/min. Labs are obtained with the following results:

Hemoglobin A1c: 6.7% Glucose: 120 mg/dl Cholesterol (plasma): 190 mg/dL Urine albumin: 60mg/24hr

Which of the following treatments is effective in slowing the progression of the most likely cause of this patient's abnormal albumin result?

Answer Choices:

A. No effective treatments B. Enalapril C. Simvastatin D. Aspirin

Correct Answer: B. Enalapril

Question 2

Clinical Scenario:

A 37-year-old man presents to the emergency department after he cut his hand while working on his car. The patient has a past medical history of antisocial personality disorder and has been incarcerated multiple times. His vitals are within normal limits. Physical exam is notable for a man covered in tattoos with many bruises over his face and torso. Inspection of the patient's right hand reveals 2 deep lacerations on the dorsal aspects of the second and third metacarpophalangeal (MCP) joints. The patient is given a tetanus vaccination, and the wound is irrigated. Which of the following is appropriate management for this patient?

Answer Choices:

A. Ciprofloxacin and topical erythromycin B. Closure of the wound with sutures C. No further management necessary D. Surgical irrigation, debridement, and amoxicillin-clavulanic acid

Correct Answer: D. Surgical irrigation, debridement, and amoxicillin-clavulanic acid

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Question 3

Clinical Scenario:

An investigator is studying the function of the lateral nucleus of the hypothalamus in an experimental animal. Using a viral vector, the genes encoding chloride-conducting channelrhodopsins are injected into this nucleus. Photostimulation of the channels causes complete inhibition of action potential generation. Persistent photostimulation is most likely to result in which of the following abnormalities in these animals?

Answer Choices:

A. Hypothermia B. Hyperthermia C. Polydipsia D. Anorexia

Correct Answer: D. Anorexia

Question 4

Clinical Scenario:

A 62-year-old man comes to the physician for a follow-up examination. For the past year, he has had increasing calf cramping in both legs when walking, especially on an incline. He has hypertension. Since the last visit 6 months ago, he has been exercising on a treadmill four times a week; he has been walking until the pain starts and then continues after a short break. He has a history of hypertension controlled with enalapril. He had smoked 2 packs of cigarettes daily for 35 years but quit 5 months ago. His temperature is 37°C (98.6°F), pulse is 84/min, and blood pressure is 132/78 mm Hg. Cardiopulmonary examination shows no abnormalities. The calves and feet are pale. Femoral pulses can be palpated bilaterally; pedal pulses are absent. His ankle-brachial index is 0.6. Which of the following is the most appropriate next step in management?

Answer Choices:

A. Clopidogrel and simvastatin B. Operative vascular reconstruction C. Percutaneous transluminal angioplasty and stenting D. Vancomycin and piperacillin

Correct Answer: A. Clopidogrel and simvastatin

Question 5

Clinical Scenario:

A group of medical students is studying bacteria and their pathogenesis. They have identified that a substantial number of bacteria cause human disease by producing exotoxins. Exotoxins are typically proteins, but they have different mechanisms of action and act at different sites. The following is a list of exotoxins together with mechanisms of action. Which of the following pairs is correctly matched?

Answer Choices:

A. Diphtheria toxin - cleaves synaptobrevin, blocking vesicle formation and the release of acetylcholine B. Cholera toxin - ADP-ribosylates Gs, keeping adenylate cyclase active and ↑ [cAMP] C. Botulinum toxin - cleaves synaptobrevin, blocking vesicle formation and the release of the inhibitory neurotransmitters GABA and glycine D. Anthrax toxin - ADP-ribosylates elongation factor - 2 (EF-2) and inhibits protein synthesis

Correct Answer: B. Cholera toxin - ADP-ribosylates Gs, keeping adenylate cyclase active and ↑ [cAMP]

Question 6

Clinical Scenario:

A 76-year-old man is brought to the emergency department by his daughter because he has been feeling lightheaded and almost passed out during dinner. Furthermore, over the past few days he has been experiencing heart palpitations. His medical history is significant for well-controlled hypertension and diabetes. Given this presentation, an electrocardiogram is performed showing an irregularly irregular tachyarrhythmia with narrow QRS complexes. The patient is prescribed a drug that decreases the slope of phase 0 of the ventricular action potential but does not change the overall duration of the action potential. Which of the following drugs is consistent with this mechanism of action?

Answer Choices:

A. Mexiletine B. Procainamide C. Propafenone D. Propanolol

Correct Answer: C. Propafenone

Question 7

Clinical Scenario:

A 60-year-old female presents to her gynecologist with bloating, abdominal discomfort, and fatigue. She has a history of hypertension and takes hydrochlorothiazide. Physical exam reveals ascites and right adnexal tenderness. Initial imaging reveals a mass in the right ovary and eventual biopsy of the mass reveals ovarian serous cystadenocarcinoma. She is started on a chemotherapeutic agent with plans for surgical resection. Soon after starting the medication, she develops dysuria and hematuria. Laboratory analysis of her urine is notable for the presence of a cytotoxic metabolite. Which of the following mechanisms of action is consistent with the medication in question?

Answer Choices:

A. DNA alkylating agent B. Platinum-based DNA intercalator C. Folate analog D. BRAF inhibitor

Correct Answer: A. DNA alkylating agent

Question 8

Clinical Scenario:

Four weeks after starting hydrochlorothiazide, a 49-year-old man with hypertension comes to the physician because of muscle cramps and weakness. His home medications also include amlodipine. His blood pressure today is 176/87 mm Hg. Physical examination shows no abnormalities. The precordial leads of a 12-lead ECG are shown. The addition of which of the following is most likely to have prevented this patient's condition?

Answer Choices:

A. Torsemide B. Nifedipine C. Eplerenone D. Hydralazine

Correct Answer: C. Eplerenone

Question 9

Clinical Scenario:

A 42-year-old male presents to your office with cellulitis on his leg secondary to a dog bite. You suspect that the causative agent is a small, facultatively anaerobic, Gram-negative rod sensitive to penicillin with clavulanate. When you ask the patient how the bite occurred, the patient explains that he had a fight with his wife earlier in the day. Frustrated with his wife, he yelled at the family pet, who bit him on the leg. Which of the following defense mechanisms was this patient employing at the time of his injury?

Answer Choices:

A. Projection B. Reaction formation C. Regression D. Displacement

Correct Answer: D. Displacement

Question 10

Clinical Scenario:

A 19-year-old man with a past medical history significant only for moderate facial acne and mild asthma presents to his primary care physician with a new rash. He notes it has developed primarily over the backs of his elbows and is itchy. He also reports a 6-month history of foul-smelling diarrhea. He has no significant social or family history. The patient's blood pressure is 109/82 mm Hg, pulse is 66/min, respiratory rate is 16/min, and temperature is 36.7°C (98.0°F). Physical examination reveals crusting vesicular clusters on his elbows with a base of erythema and edema. What is the most likely underlying condition?

Answer Choices:

A. Food allergy B. Type 2 diabetes mellitus C. Celiac disease D. IgA nephropathy

Correct Answer: C. Celiac disease

Question 11

Clinical Scenario:

A 64-year-old male presents to the emergency room with difficulty breathing. He recently returned to the USA following a trip to Singapore. He reports that he developed pleuritic chest pain, shortness of breath, and a cough. His temperature is 99°F (37.2°C), blood pressure is 140/85 mmHg, pulse is 110/min, and respirations are 24/min. A spiral CT reveals a pulmonary embolus in the right segmental pulmonary artery. Results from a complete blood count are all within normal limits. He is admitted and started on unfractionated heparin. Four days later, the patient develops unprovoked epistaxis. A complete blood count reveals the following:

Leukocyte count: 7,000/mm^3 Hemoglobin: 14 g/dl Hematocrit: 44% Platelet count 40,000/mm^3

What is the underlying pathogenesis of this patient’s condition?

Answer Choices:

A. Loss of vitamin K-dependent clotting factors B. Autoantibodies directed against platelet factor 4 C. Medication-mediated platelet aggregation D. ADAMTS13 deficiency

Correct Answer: B. Autoantibodies directed against platelet factor 4

Question 12

Clinical Scenario:

A 16-year-old boy comes to the emergency department because of painful urination and urethral discharge for 3 days. He has multiple sexual partners and only occasionally uses condoms. His vital signs are within normal limits. The result of nucleic acid amplification testing for Neisseria gonorrhoeae is positive. The patient requests that his parents not be informed of the diagnosis. Which of the following initial actions by the physician is most appropriate?

Answer Choices:

A. Perform urethral swab culture for antibiotic sensitivities B. Request parental consent prior to prescribing antibiotics C. Discuss results with patient's primary care physician D. Administer intramuscular and oral antibiotics

Correct Answer: D. Administer intramuscular and oral antibiotics

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Question 13

Clinical Scenario:

A 5-year-old girl is brought to the physician by her mother because of a 3-week history of a foul-smelling discharge from the left nostril. There was one episode of blood-tinged fluid draining from the nostril during this period. She has been mouth-breathing in her sleep for the past 4 days. She was born at term. Her 1-year-old brother was treated for viral gastroenteritis 3 weeks ago. She is at 60th percentile for height and at 70th percentile for weight. Her temperature is 37°C (98.6°F), pulse is 96/min, respirations are 23/min, and blood pressure is 96/54 mm Hg. Examination shows mucopurulent discharge in the left nasal cavity. Oral and otoscopic examination is unremarkable. Endoscopic examination of the nose confirms the diagnosis. Which of the following is the most appropriate next step in management?

Answer Choices:

A. Transnasal puncture and stenting B. Foreign body extraction C. Adenoidectomy D. Intranasal glucocorticoid therapy "

Correct Answer: B. Foreign body extraction

Question 14

Clinical Scenario:

A 39-year-old woman comes to the physician because of recurrent episodes of severe pain over her neck, back, and shoulders for the past year. The pain worsens with exercise and lack of sleep. Use of over-the-counter analgesics have not resolved her symptoms. She also has stiffness of the shoulders and knees and tingling in her upper extremities that is worse in the morning. She takes escitalopram for generalized anxiety disorder. She also has tension headaches several times a month. Her maternal uncle has ankylosing spondylitis. Examination shows marked tenderness over the posterior neck, bilateral mid trapezius, and medial aspect of the left knee. Muscle strength is normal. Laboratory studies, including a complete blood count, erythrocyte sedimentation rate, and thyroid-stimulating hormone are within the reference ranges. X-rays of her cervical and lumbar spine show no abnormalities. Which of the following is the most likely diagnosis?

Answer Choices:

A. Polymyalgia rheumatica B. Fibromyalgia C. Polymyositis D. Major depressive disorder

Correct Answer: B. Fibromyalgia

Question 15

Clinical Scenario:

A 63-year-old man comes to the physician because of a 2-day history of fever and blood-tinged sputum. He has also had a productive cough for 1 year and has had 3 episodes of sinusitis during this time. Physical examination shows palpable erythematous skin lesions over his hands and feet that do not blanch on pressure. There are ulcerations of the nasopharyngeal mucosa and a perforation of the nasal septum. His serum creatinine is 2.6 mg/dL. Urinalysis shows acanthocytes, 70 RBCs/hpf, 2+ proteinuria, and RBC casts. An x-ray of the chest shows multiple, cavitating, nodular lesions bilaterally. Further evaluation of this patient is most likely to show which of the following findings?

Answer Choices:

A. Elevated anti-Smith titers B. Elevated serum IgA titers C. Positive tuberculin test D. Elevated c-ANCA titers

Correct Answer: D. Elevated c-ANCA titers

Question 16

Clinical Scenario:

A 26-year-old medical student comes to the physician for a chest x-ray to rule out active pulmonary tuberculosis. He needs a medical and radiological report before starting a medical internship in South Africa. He has no history of serious illness and does not complain of any symptoms. He has smoked 1 pack of cigarettes daily for the past 6 years. He does not drink alcohol. He is 190 cm (6 ft 3 in) tall and weighs 75 kg (165 lbs); BMI is 20.8 kg/m2. His temperature is 37°C (98.6°F), pulse is 80/min, respirations are 18/min, and blood pressure is 128/89 mm Hg. The lungs are clear to auscultation. Cardiac examination shows no abnormalities. The x-ray of the chest shows a small pneumothorax (rim of < 2 cm) between the upper left lung margin and the chest wall. Which of the following is the most appropriate next step in management of this patient?

Answer Choices:

A. Emergent needle thoracostomy B. Observation and follow-up x-ray C. Immediate intubation and assisted ventilation D. Urgent chest tube placement

Correct Answer: B. Observation and follow-up x-ray

Question 17

Clinical Scenario:

A 33-year-old female presents to her primary care physician complaining of heat intolerance and difficulty sleeping over a one month period. She also reports that she has lost 10 pounds despite no changes in her diet or exercise pattern. More recently, she has developed occasional unprovoked chest pain and palpitations. Physical examination reveals a nontender, mildly enlarged thyroid gland. Her patellar reflexes are 3+ bilaterally. Her temperature is 99°F (37.2°C), blood pressure is 135/85 mmHg, pulse is 105/min, and respirations are 18/min. Laboratory analysis is notable for decreased TSH. Which of the following pathophysiologic mechanisms contributed to the cardiovascular symptoms seen in this patient?

Answer Choices:

A. Increased numbers of a1-adrenergic receptors B. Decreased numbers of a1-adrenergic receptors C. Decreased numbers of a2-adrenergic receptors D. Increased sensitivity of ß1-adrenergic receptors

Correct Answer: D. Increased sensitivity of ß1-adrenergic receptors

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Question 18

Clinical Scenario:

An epidemiologist is interested in studying the clinical utility of a free computerized social skills training program for children with autism. A total of 125 participants with autism (mean age: 12 years) were recruited for the study and took part in weekly social skills training sessions for 3 months. Participants were recruited from support groups in a large Northeastern US city for parents with autistic children. Parents in the support group were very eager to volunteer for the study, and over 300 children were placed on a waiting list while the study was conducted. At baseline and at the end of the 3-month period, participants were observed during a videotaped social play exercise and scored on a social interaction rating scale by their parents. Social interaction rating scores following the 3-month intervention were more than twice as high as baseline scores ($p < 0.001$). During exit interviews, one parent commented, ""I knew from the start that this program was going to be life-changing for my son!"" This sentiment was echoed by a number of other parents. Which of the following is the most likely explanations for the study's result?"

Answer Choices:

A. Social desirability bias B. Observer bias C. Sampling bias D. Confounding bias

Correct Answer: B. Observer bias

Question 19

Clinical Scenario:

A 55-year-old man comes to the physician because of difficulties achieving an erection for the past year. A medication is prescribed that inhibits cyclic GMP phosphodiesterase type 5. Which of the following is the most likely site of action of the prescribed drug?

Answer Choices:

A. Corpus cavernosum B. Prostate smooth muscle C. Corpus spongiosum D. Pudendal nerve

Correct Answer: A. Corpus cavernosum

Question 20

Clinical Scenario:

A 35-year-old woman has been trying to conceive with her 37-year-old husband for the past 4 years. After repeated visits to a fertility clinic, she finally gets pregnant. Although she missed most of her antenatal visits, her pregnancy was uneventful. A baby girl is born at the 38th week of gestation with some abnormalities. She has a flat face with upward-slanting eyes and a short neck. The tongue seems to be protruding from a small mouth. She has poor muscle tone and excessive joint laxity. The pediatrician orders an analysis of the infant's chromosomes, also known as a karyotype (see image). The infant is most likely to suffer from which of the following conditions in the future?

Answer Choices:

A. Acute lymphoblastic leukemia B. Chronic myelogenous leukemia C. Immotile cilia syndrome D. Macroorchidism

Correct Answer: A. Acute lymphoblastic leukemia

Question 21

Clinical Scenario:

A researcher measures action potential propagation velocity in various regions of the heart in a 42-year-old Caucasian female. Which of the following set of measurements corresponds to the velocities found in the atrial muscle, AV Node, Purkinje system, and venticular muscle, respectively?

Answer Choices:

A. 2.2 m/s, 0.3 m/s, 0.05 m/s, 1.1 m/s B. 1.1 m/s, 0.05 m/s, 2.2 m/s, 0.3 m/s C. 0.5 m/s, 1.1 m/s, 2.2 m/s, 3 m/s D. 0.05 m/s, 1.1 m/s, 2.2 m/s, 3.3 m/s

Correct Answer: B. 1.1 m/s, 0.05 m/s, 2.2 m/s, 0.3 m/s

Question 22

Clinical Scenario:

A male newborn is delivered at term to a 30-year-old woman. Pregnancy and delivery were uncomplicated. At birth, the umbilical cord is noted to be large. When the newborn cries, straw-colored fluid leaks from the umbilicus. The external genitalia appear normal. Which of the following is the most likely cause of this newborn's symptoms?

Answer Choices:

A. Abnormal fusion of the urethral folds B. Failed closure of the vitelline duct C. Failed obliteration of an allantoic remnant D. Infection of the umbilical cord stump

Correct Answer: C. Failed obliteration of an allantoic remnant

Question 23

Clinical Scenario:

A 7-year-old boy is brought to the pediatrician by his parents for concern of general fatigue and recurrent abdominal pain. You learn that his medical history is otherwise unremarkable and that these symptoms started about 3 months ago after they moved to a different house. Based on clinical suspicion labs are obtained that reveal a microcytic anemia with high-normal levels of ferritin. Examination of a peripheral blood smear shows findings that are demonstrated in the figure provided. Which of the following is the most likely mechanism responsible for the anemia in this patient?

Answer Choices:

A. Chronic loss of blood through GI tract B. X-linked mutation of ALA synthetase C. Inflammation due to occult abdominal malignancy D. Inhibition of ALA dehydratase and ferrochelatase

Correct Answer: D. Inhibition of ALA dehydratase and ferrochelatase

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Question 24

Clinical Scenario:

A previously healthy 36-year-old man comes to the physician for a yellow discoloration of his skin and dark-colored urine for 2 weeks. He does not drink any alcohol. Physical examination shows jaundice. Abdominal and neurologic examinations show no abnormalities. Serum studies show increased levels of alanine aminotransferase (ALT) and aspartate aminotransferase (AST). A liver biopsy is performed and a photomicrograph after periodic acid-Schiff-staining is shown. Which of the following is the most likely additional finding in this patient?

Answer Choices:

A. Bullous changes of the lung bases on chest CT B. Beading of intra- and extrahepatic bile ducts on ERCP C. Myocardial iron deposition on cardiovascular MRI D. Dark corneal ring on slit-lamp examination

Correct Answer: A. Bullous changes of the lung bases on chest CT

Question 25

Clinical Scenario:

A 56-year-old man presents with feelings of anxiety and fatigue for the past 4 months. He says that he has also had some weight loss, as well as occasional double vision and a gritty sensation in his eyes for the last 2 months, which is worse at the end of the day. He has also noticed some painless swelling in his fingers and lower legs during the same time period. The patient denies any recent history of fevers, chills, night sweats, nausea, or vomiting. Current medications include aspirin, simvastatin, and omeprazole. Which of the following mechanisms is most likely responsible for this patient’s condition?

Answer Choices:

A. Autoantibodies resulting in tissue destruction B. Autoantibody stimulation of a receptor C. Excessive exogenous hormone use D. Infiltration of tissue by neoplastic cells

Correct Answer: B. Autoantibody stimulation of a receptor

Question 26

Clinical Scenario:

A 36-year-old primigravid woman at 22 weeks' gestation comes to the physician for a routine prenatal visit. Her previous prenatal visits showed no abnormalities. She has hyperthyroidism treated with methimazole. She previously smoked one pack of cigarettes daily for 15 years but quit 6 years ago. She reports gaining weight after quitting smoking, after which she developed her own weight loss program. She is 168 cm (5 ft 6 in) tall and weighs 51.2 kg (112.9 lb); BMI is 18.1 kg/m2. Her temperature is 37°C (98.5°F), pulse is 88/min, and blood pressure is 115/72 mm Hg. Pelvic examination shows no abnormalities. The fundus is palpated between the symphysis and the umbilicus. Ultrasound shows a fetal head at the 20th percentile and the abdomen at the 9th percentile. Fetal birth weight is estimated at the 9th percentile and a decreased amniotic fluid index is noted. The maternal quadruple screening test was normal. Thyroid-stimulating hormone is 0.4 mIU/mL, triiodothyronine (T3) is 180 ng/dL, and thyroxine (T4) is 10 μg/dL. Which of the following is the strongest predisposing factor for the ultrasound findings in this patient?

Answer Choices:

A. Maternal malnutrition B. Advanced maternal age C. Fetal aneuploidy D. History of tobacco use

Correct Answer: A. Maternal malnutrition

Question 27

Clinical Scenario:

A 28-year-old woman is brought to the emergency department by her friends. She is naked except for a blanket and speaking rapidly and incoherently. Her friends say that she was found watering her garden naked and refused to put on any clothes when they tried to make her do so, saying that she has accepted how beautiful she is inside and out. Her friends say she has also purchased a new car she can not afford. They are concerned about her, as they have never seen her behave this way before. For the past week, she has not shown up at work and has been acting ‘strangely’. They say she was extremely excited and has been calling them at odd hours of the night to tell them about her future plans. Which of the following drug mechanisms will help with the long-term management this patient’s symptoms?

Answer Choices:

A. Inhibit the reuptake norepinephrine and serotonin from the presynaptic cleft B. Inhibition of inositol monophosphatase and inositol polyphosphate 1-phosphatase C. Increase the concentration of dopamine and norepinephrine at the synaptic cleft D. Modulate the activity of ϒ-aminobutyric acid receptors

Correct Answer: B. Inhibition of inositol monophosphatase and inositol polyphosphate 1-phosphatase

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Question 28

Clinical Scenario:

A 58-year-old man with liver cirrhosis presents to his primary care physician complaining of increased abdominal girth and early satiety. He drinks 2–4 glasses of wine with dinner and recalls having had abnormal liver enzymes in the past. Vital signs include a temperature of 37.1°C (98.7°F), blood pressure of 110/70 mm Hg, and a pulse of 75/min. Physical examination reveals telangiectasias, mild splenomegaly, palpable firm liver, and shifting dullness. Liver function is shown: Total bilirubin 3 mg/dL Aspartate aminotransferase (AST) 150 U/L Alanine aminotransferase (ALT) 70 U/L Total albumin 2.5 g/dL Abdominal ultrasonography confirms the presence of ascites. Diagnostic paracentesis is performed and its results are shown: Polymorphonuclear cell count 10 cells/mm Ascitic protein 1 g/dL Which of the following best represent the mechanism of ascites in this patients?

Answer Choices:

A. Peritoneal carcinomatosis B. Serositis C. High sinusoidal pressure D. Pancreatic disease

Correct Answer: C. High sinusoidal pressure

Question 29

Clinical Scenario:

A molecular biologist is studying the roles of different types of ion channels regulating cardiac excitation. He identifies a voltage-gated calcium channel in the sinoatrial node, which is also present throughout the myocardium. The channel is activated at ~ -40 mV of membrane potential, undergoes voltage-dependent inactivation, and is highly sensitive to nifedipine. Which of the following phases of the action potential in the sinoatrial node is primarily mediated by ion currents through the channel that the molecular biologist is studying?

Answer Choices:

A. Phase 0 B. Phase 1 C. Phase 3 D. Phase 4

Correct Answer: A. Phase 0

Question 30

Clinical Scenario:

A healthy 36-year-old Caucasian man takes part in an experimental drug trial. The drug is designed to lower glomerular filtration rate (GFR) while simultaneously raising the filtration fraction. Which of the following effects on the glomerulus would you expect the drug to have?

Answer Choices:

A. Afferent arteriole constriction and efferent arteriole vasodilation B. Afferent arteriole constriction and efferent arteriole constriction C. Afferent arteriole dilation and efferent arteriole constriction D. Increased oncotic pressure in Bowman's space

Correct Answer: B. Afferent arteriole constriction and efferent arteriole constriction

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Question 31

Clinical Scenario:

A 16-year-old boy is brought in to a psychiatrist's office by his mother for increasingly concerning erratic behavior. Her son has recently entered a new relationship, and he constantly voices beliefs that his girlfriend is cheating on him. He ended his last relationship after voicing the same beliefs about his last partner. During the visit, the patient reports that these beliefs are justified, since everyone at school is “out to get him.” He says that even his teachers are against him, based on their criticism of his schoolwork. His mother adds that her son has always held grudges against people and has always taken comments very personally. The patient has no psychiatric history and is in otherwise good health. What condition is this patient genetically predisposed for?

Answer Choices:

A. Major depressive disorder B. Narcolepsy C. Schizophrenia D. Substance use disorder

Correct Answer: C. Schizophrenia

Question 32

Clinical Scenario:

A researcher is studying the mammalian immune response with an unknown virus. A group of mice are inoculated with the virus, and blood is subsequently drawn from these animals at various intervals to check immunoglobulin levels. Which of the following is a critical step in the endogenous pathway of antigen presentation for the virus model presented above?

Answer Choices:

A. Degradation of the antigen by the proteases in the phagolysosome B. Translocation of the antigen into the endoplasmic reticulum via TAP proteins C. Binding of the peptide to MHC class II D. Interaction of the MHC class II complex with its target CD4+ T cell

Correct Answer: B. Translocation of the antigen into the endoplasmic reticulum via TAP proteins

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Question 33

Clinical Scenario:

An 8-year-old boy is brought to the physician by his parents because of repeated episodes of “daydreaming.” The mother reports that during these episodes the boy interrupts his current activity and just “stares into space.” She says that he sometimes also smacks his lips. The episodes typically last 1–2 minutes. Over the past 2 months, they have occurred 2–3 times per week. The episodes initially only occurred at school, but last week the patient had one while he was playing baseball with his father. When his father tried to talk to him, he did not seem to listen. After the episode, he was confused for 10 minutes and too tired to play. The patient has been healthy except for an episode of otitis media 1 year ago that was treated with amoxicillin. Vital signs are within normal limits. Physical and neurological examinations show no other abnormalities. Further evaluation of this patient is most likely to show which of the following findings?

Answer Choices:

A. Defiant behavior towards figures of authority B. Impairment in communication and social interaction C. Temporal lobe spikes on EEG D. Conductive hearing loss on audiometry

Correct Answer: C. Temporal lobe spikes on EEG

Question 34

Clinical Scenario:

A 49-year-old man with a past medical history of hypertension on amlodipine presents to your office to discuss ways to lessen his risk of complications from heart disease. After a long discussion, he decides to significantly decrease his intake of trans fats in an attempt to lower his risk of coronary artery disease. Which type of prevention is this patient initiating?

Answer Choices:

A. Primary prevention B. Secondary prevention C. Tertiary prevention D. Delayed prevention

Correct Answer: A. Primary prevention

Question 35

Clinical Scenario:

A newborn is evaluated by the on-call pediatrician. She was born at 33 weeks gestation via spontaneous vaginal delivery to a 34-year-old G1P1. The pregnancy was complicated by poorly controlled diabetes mellitus type 2. Her birth weight was 3,700 g and the appearance, pulse, grimace, activity, and respiration (APGAR) scores were 7 and 8 at 1 and 5 minutes, respectively. The umbilical cord had 3 vessels and the placenta was tan-red with all cotyledons intact. Fetal membranes were tan-white and semi-translucent. The normal-appearing placenta and cord were sent to pathology for further evaluation. On physical exam, the newborn’s vital signs include: temperature 36.8°C (98.2°F), blood pressure 60/44 mm Hg, pulse 185/min, and respiratory rate 74/min. She presents with nasal flaring, subcostal retractions, and mild cyanosis. Breath sounds are decreased at the bases of both lungs. Arterial blood gas results include a pH of 6.91, partial pressure of carbon dioxide ($PaCO_2$) 97 mm Hg, partial pressure of oxygen (PaO_2) 25 mm Hg, and base excess of 15.5 mmol/L (reference range: ± 3 mmol/L). What is the most likely diagnosis?

Answer Choices:

A. Transient tachypnea of the newborn B. Infant respiratory distress syndrome C. Meconium aspiration syndrome D. Fetal alcohol syndrome

Correct Answer: B. Infant respiratory distress syndrome

Question 36

Clinical Scenario:

A 28-year-old man presents to the emergency department with vomiting. He states that he has experienced severe vomiting starting last night that has not been improving. He states that his symptoms improve with hot showers. The patient has presented to the emergency department with a similar complaints several times in the past as well as for intravenous drug abuse. His temperature is 99.5°F (37.5°C), blood pressure is 127/68 mmHg, pulse is 110/min, respirations are 22/min, and oxygen saturation is 98% on room air. Physical exam is deferred as the patient is actively vomiting. Which of the following is associated with the most likely diagnosis?

Answer Choices:

A. Alcohol use B. Marijuana use C. Substance withdrawal D. Viral gastroenteritis

Correct Answer: B. Marijuana use

Question 37

Clinical Scenario:

A 5-year-old boy presents to his pediatrician for a well-child visit. His mother reports him to be doing well and has no concerns. The boy was born at 39 weeks gestation via spontaneous vaginal delivery. He is up to date on all vaccines and is meeting all developmental milestones. On physical exam, he is noted to have a right upper extremity blood pressure of 150/80 mm Hg. 2+ radial pulses and trace femoral pulses are felt. Cardiac auscultation reveals a regular rate and rhythm with a normal S1 and S2. A 2/6 long systolic murmur with systolic ejection click is heard over left sternal border and back. The point of maximal impact is normal Which of the following is the most likely diagnosis?

Answer Choices:

A. Takayasu arteritis B. Interrupted aortic arch C. Pheochromocytoma D. Coarctation of the aorta

Correct Answer: D. Coarctation of the aorta

Question 38

Clinical Scenario:

A senior medicine resident receives negative feedback on a grand rounds presentation from his attending. He is told sternly that he must improve his performance on the next project. Later that day, he yells at his medical student for not showing enough initiative, though he had voiced only satisfaction with the student's performance up until this point. Which of the following psychological defense mechanisms is he demonstrating?

Answer Choices:

A. Countertransference B. Externalization C. Displacement D. Projection "

Correct Answer: C. Displacement

Question 39

Clinical Scenario:

A 23-year-old man is brought to the emergency department because of severe right shoulder pain and inability to move the shoulder for the past 30 minutes. The pain began after being tackled while playing football. He has nausea but has not vomited. He is in no apparent distress. Examination shows the right upper extremity externally rotated and slightly abducted. Palpation of the right shoulder joint shows tenderness and an empty glenoid fossa. The right humeral head is palpated below the coracoid process. The left upper extremity is unremarkable. The radial pulses are palpable bilaterally. Which of the following is the most appropriate next step in management?

Answer Choices:

A. Neer impingement test B. Closed reduction C. Test sensation of the lateral shoulder D. Drop arm test

Correct Answer: C. Test sensation of the lateral shoulder

Question 40

Clinical Scenario:

A researcher is studying the interactions between foreign antigens and human immune cells. She has isolated a line of lymphocytes that is known to bind antigen-presenting cells. From this cell line, she has isolated a cell surface protein that binds the constant portion of the class I major histocompatibility complex molecule. The activation of this specific cell line requires co-activation via which of the following signaling molecules?

Answer Choices:

A. Interleukin 2 B. Interleukin 4 C. Interleukin 6 D. Interleukin 8

Correct Answer: A. Interleukin 2

Question 41

Clinical Scenario:

A 33-year-old man comes to the physician with his wife for evaluation of infertility. They have been unable to conceive for 2 years. The man reports normal libido and erectile function. He has smoked one pack of cigarettes daily for 13 years. He does not take any medications. He has a history of right-sided cryptorchidism that was surgically corrected when he was 7 years of age. Physical examination shows no abnormalities. Analysis of his semen shows a low sperm count. Laboratory studies are most likely to show which of the following?

Answer Choices:

A. Increased placental ALP concentration B. Increased prolactin concentration C. Decreased inhibin B concentration D. Decreased FSH concentration

Correct Answer: C. Decreased inhibin B concentration

Question 42

Clinical Scenario:

A 72-year-old man comes to the physician because of a 6-month history of intermittent dull abdominal pain that radiates to the back. He has smoked one pack of cigarettes daily for 50 years. His blood pressure is 145/80 mm Hg. Abdominal examination shows generalized tenderness and a pulsatile mass in the periumbilical region on deep palpation. Further evaluation of the affected blood vessel is most likely to show which of the following?

Answer Choices:

A. Accumulation of foam cells in the tunica intima B. Obliterative inflammation of the vasa vasorum C. Necrotizing inflammation of the entire vessel wall D. Fragmentation of elastic tissue in the tunica media

Correct Answer: A. Accumulation of foam cells in the tunica intima

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Question 43

Clinical Scenario:

A 43-year-old man comes to the physician because of left flank pain and nausea for 2 hours. The pain comes in waves and radiates to his groin. Over the past year, he has had intermittent pain in the bilateral flanks and recurrent joint pain in the toes, ankles, and fingers. He has not seen a physician in over 10 years. He takes no medications. He drinks 3–5 beers daily. His sister has rheumatoid arthritis. Vital signs are within normal limits. Physical examination shows marked tenderness bilaterally in the costovertebral areas. A photograph of the patient's left ear is shown. A CT scan of the abdomen shows multiple small kidney stones and a 7-mm left distal ureteral stone. A biopsy of the patient's external ear findings is most likely to show which of the following?

Answer Choices:

A. Cholesterol B. Ammonium magnesium phosphate C. Monosodium urate D. Calcium oxalate

Correct Answer: C. Monosodium urate

Question 44

Clinical Scenario:

An 18-month-old boy presents to the clinic with his mother for evaluation of a rash around the eyes and mouth. His mother states that the rash appeared 2 weeks ago and seems to be very itchy because the boy scratches his eyes often. The patient is up to date on all of his vaccinations and is meeting all developmental milestones. He has a history of asthma that was recently diagnosed. On examination, the patient is playful and alert. He has scaly, erythematous skin surrounding both eyes and his mouth. Bilateral pupils are equal and reactive to light and accommodation, and conjunctiva is clear, with no evidence of jaundice or exudates. The pharynx and oral mucosa are within normal limits, and no lesions are present. Expiratory wheezes can be heard in the lower lung fields bilaterally. What is this most likely diagnosis in this patient?

Answer Choices:

A. Viral conjunctivitis B. Impetigo C. Atopic dermatitis D. Scalded skin syndrome

Correct Answer: C. Atopic dermatitis

Question 45

Clinical Scenario:

A scientist is researching the long term effects of the hepatitis viruses on hepatic tissue. She finds that certain strains are oncogenic and increase the risk of hepatocellular carcinoma. However, they appear to do so via different mechanisms. Which of the following answer choices correctly pairs the hepatitis virus with the correct oncogenic process?

Answer Choices:

A. Hepatitis A virus - chronic inflammation B. Hepatitis A virus - integration of viral DNA into host hepatocyte genome C. Hepatitis B virus - integration of viral DNA into host hepatocyte genome D. Hepatitis E virus - integration of viral DNA into host hepatocyte genome

Correct Answer: C. Hepatitis B virus - integration of viral DNA into host hepatocyte genome

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Question 46

Clinical Scenario:

A 56-year-old man with chronic kidney failure is brought to to the emergency department by ambulance after he passed out during dinner. On presentation, he is alert and complains of shortness of breath as well as chest palpitations. An EKG is obtained demonstrating an irregular rhythm consisting of QT amplitudes that vary in height over time. Other findings include uncontrolled contractions of his muscles. Tapping of his cheek does not elicit any response. Over-repletion of the serum abnormality in this case may lead to which of the following?

Answer Choices:

A. Bradycardia B. Diffuse calcifications C. Kidney stones D. Seizures

Correct Answer: A. Bradycardia

Question 47

Clinical Scenario:

A 22-year-old soldier sustains a gunshot wound to the left side of the chest during a deployment in Syria. The soldier and her unit take cover from gunfire in a nearby farmhouse, and a combat medic conducts a primary survey of her injuries. She is breathing spontaneously. Two minutes after sustaining the injury, she develops severe respiratory distress. On examination, she is agitated and tachypneic. There is an entrance wound at the midclavicular line at the 2nd rib and an exit wound at the left axillary line at the 4th rib. There is crepitus on the left side of the chest wall. Which of the following is the most appropriate next step in management?

Answer Choices:

A. Endotracheal intubation B. Intravenous administration of fentanyl C. Ultrasonography of the chest D. Needle thoracostomy "

Correct Answer: D. Needle thoracostomy "

Question 48

Clinical Scenario:

A 63-year-old man presents to the emergency department complaining of sudden-onset severe dyspnea and right-sided chest pain. The patient has a history of chronic obstructive pulmonary disease, hypertension, peptic ulcer disease, and hyperthyroidism. He has smoked a pack of cigarettes daily for 20 years, drinks socially, and does not take illicit drugs. The blood pressure is 130/80 mm Hg, the pulse is 98/min and regular, and the respiratory rate is 20/min. Pulse oximetry shows 90% on room air. On physical examination, he is in mild respiratory distress. Tactile fremitus and breath sounds are decreased on the right, with hyperresonance on percussion. The trachea is midline and no heart murmurs are heard. Which of the following is the most likely underlying mechanism of this patient's current condition?

Answer Choices:

A. Compression of a main bronchus due to neoplasia B. Formation of an intimal flap in the aorta C. Increased myocardial oxygen demand D. Rupture of an apical alveolar bleb

Correct Answer: D. Rupture of an apical alveolar bleb

Question 49

Clinical Scenario:

An investigator is studying mechanisms of urea excretion in humans. During the experiment, a healthy male volunteer receives a continuous infusion of para-aminohippurate (PAH) to achieve a PAH plasma concentration of 0.01 mg/mL. A volume of 1.0 L of urine is collected over a period of 10 hours; the urine flow rate is 1.66 mL/min. The urinary concentration of PAH is measured to be 3.74 mg/mL and his serum concentration of urea is 0.2 mg/mL. Assuming a normal filtration fraction of 20%, which of the following best estimates the filtered load of urea in this patient?

Answer Choices:

A. 25 mg/min B. 124 mg/min C. 7 mg/min D. 166 mg/min

Correct Answer: A. 25 mg/min

Question 50

Clinical Scenario:

A 20-year-old man who is a biology major presents to his physician for a simple check-up. He is informed that he hasn’t received a hepatitis B vaccine. When the first injection is applied, the medical professional informs him that he will need to come back 2 more times on assigned days, since the vaccine is given in 3 doses. Which of the following antibodies is the physician trying to increase in the college student as a result of the first vaccination?

Answer Choices:

A. IgA B. IgM C. IgD D. IgE

Correct Answer: B. IgM

Question 51

Clinical Scenario:

An investigator is studying the mechanism of HIV infection in cells obtained from a human donor. The effect of a drug that impairs viral fusion and entry is being evaluated. This drug acts on a protein that is cleaved off of a larger glycosylated protein in the endoplasmic reticulum of the host cell. The protein that is affected by the drug is most likely encoded by which of the following genes?

Answer Choices:

A. rev B. gag C. env D. tat

Correct Answer: C. env

Question 52

Clinical Scenario:

A 27-year old gentleman presents to the primary care physician with the chief complaint of "feeling down" for the last 6 weeks. He describes trouble falling asleep at night, decreased appetite, and recent feelings of intense guilt regarding the state of his personal relationships. He says that everything "feels slower" than it used to. He endorses having a similar four-week period of feeling this way last year. He denies thoughts of self-harm or harm of others. He also denies racing thoughts or delusions of grandeur. Which of the following would be an INAPPROPRIATE first line treatment for him?

Answer Choices:

A. Psychotherapy B. Citalopram C. Electroconvulsive therapy D. Sertraline

Correct Answer: C. Electroconvulsive therapy

Question 53

Clinical Scenario:

A 7-month-old boy presents with fever, chills, cough, runny nose, and watery eyes. His elder brother is having similar symptoms. Past medical history is unremarkable. The patient is diagnosed with an influenza virus infection. Assuming that this is the child’s first exposure to the influenza virus, which of the following immune mechanisms will most likely function to combat this infection?

Answer Choices:

A. Natural killer cell-induced lysis of virus infected cells B. Presentation of viral peptides on MHC- class I of CD4+ T cells C. Binding of virus-specific immunoglobulins to free virus D. Eosinophil-mediated lysis of virus infected cells

Correct Answer: A. Natural killer cell-induced lysis of virus infected cells

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Question 54

Clinical Scenario:

A 75-year-old woman presents to her physician with a cough and shortness of breath. She says that cough gets worse at night and her shortness of breath occurs with moderate exertion or when lying flat. She says these symptoms have been getting worse over the last 6 months. She mentions that she has to use 3 pillows while sleeping in order to relieve her symptoms. She denies any chest pain, chest tightness, or palpitations. Past medical history is significant for hypertension and diabetes mellitus type 2. Her medications are amiloride, glyburide, and metformin. Family history is significant for her father who also suffered diabetes mellitus type 2 before his death at 90 years old. The patient says she drinks alcohol occasionally but denies any smoking history. Her blood pressure is 130/95 mm Hg, temperature is 36.5°C (97.7°F), and heart rate is 100/min. On physical examination, she has a sustained apical impulse, a normal S1 and S2, and a loud S4 without murmurs. There are bilateral crackles present bilaterally. A chest radiograph shows a mildly enlarged cardiac silhouette. A transesophageal echocardiogram is performed and shows a normal left ventricular ejection fraction. Which of the following myocardial changes is most likely present in this patient?

Answer Choices:

A. Ventricular hypertrophy with sarcomeres duplicated in series B. Ventricular hypertrophy with sarcomeres duplicated in parallel C. Asymmetric hypertrophy of the interventricular septum D. Granuloma consisting of lymphocytes, plasma cells and macrophages surrounding necrotic

Correct Answer: B. Ventricular hypertrophy with sarcomeres duplicated in parallel

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Question 55

Clinical Scenario:

A 30-year-old man presents to his primary care physician for pain in his left ankle. The patient states that he was at karate practice when he suddenly felt severe pain in his ankle forcing him to stop. The patient has a past medical history notable for type I diabetes and is currently being treated for an episode of acute bacterial sinusitis with moxifloxacin. The patient recently had to have his insulin dose increased secondary to poorly controlled blood glucose levels. Otherwise, the patient takes ibuprofen for headaches and loratadine for seasonal allergies. Physical exam reveals a young healthy man in no acute distress. Pain is elicited over the Achilles tendon with dorsiflexion of the left foot. Pain is also elicited with plantar flexion of the left foot against resistance. Which of the following is the best next step in management?

Answer Choices:

A. Change antibiotics and refrain from athletic activities B. Ibuprofen and rest C. Orthopedic ankle brace D. Rehabilitation exercises and activity as tolerated

Correct Answer: A. Change antibiotics and refrain from athletic activities

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Question 56

Clinical Scenario:

A 32-year-old woman presents to her physician concerned about wet spots on the inside part of her dress shirts, which she thinks it may be coming from one of her breasts. She states that it is painless and that the discharge is usually blood-tinged. She denies any history of malignancy in her family and states that she has been having regular periods since they first started at age 13. She does not have any children. The patient has normal vitals and denies any cough, fever. On exam, there are no palpable masses, and the patient does not have any erythema or induration. What is the most likely diagnosis?

Answer Choices:

A. Paget's disease B. Breast abscess C. Ductal carcinoma D. Intraductal papilloma

Correct Answer: D. Intraductal papilloma

Question 57

Clinical Scenario:

A 17-year-old girl presents to the family doctor with fever, headache, sore throat, dry cough, myalgias, and weakness. Her symptoms began acutely 2 days ago. On presentation, her blood pressure is 110/80 mm Hg, heart rate is 86/min, respiratory rate is 18/min, and temperature is 39.0°C (102.2°F). Physical examination reveals conjunctival injection and posterior pharyngeal wall erythema. Rapid diagnostic testing of a throat swab for influenza A+B shows positive results. Which of the following statements is true regarding the process of B cell clonal selection and the formation of specific IgG antibodies against influenza virus antigens in this patient?

Answer Choices:

A. The first event that occurs after B lymphocyte activation is V(D)J recombination. B. During antibody class switching, variable region of antibody heavy chain changes, and the constant one stays the same. C. Deletions are the most common form of mutations that occur during somatic hypermutation in this patient's B cells. D. After somatic hypermutation, only a small amount of B cells antigen receptors have increased affinity for the antigen.

Correct Answer: D. After somatic hypermutation, only a small amount of B cells antigen receptors have increased affinity for the antigen.

Question 58

Clinical Scenario:

A 48-year-old female comes into the ER with chest pain. An electrocardiogram (EKG) shows a heart beat of this individual in Image A. The QR segment best correlates with what part of the action potential of the ventricular myocyte shown in Image B?

Answer Choices:

A. Phase 0, which is primarily characterized by sodium influx B. Phase 0, which is primarily characterized by potassium efflux C. Phase 1, which is primarily characterized by potassium and chloride efflux D. Phase 1, which is primarily characterized by calcium efflux

Correct Answer: A. Phase 0, which is primarily characterized by sodium influx

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Question 59

Clinical Scenario:

A 26-year-old nurse presents 12 hours after she accidentally stuck herself with a blood-contaminated needle. She reported the accident appropriately and now seeks post-exposure prophylaxis. She does not have any complaints at the moment of presentation. Her vital signs include: blood pressure 125/80 mm Hg, heart rate 71/min, respiratory rate 15/min, and temperature 36.5℃ (97.7℉). Physical examination is unremarkable. The nurse has prescribed a post-exposure prophylaxis regimen which includes tenofovir, emtricitabine, and raltegravir. How will tenofovir change the maximum reaction rate (Vm) and Michaelis constant (Km) of the viral reverse transcriptase?

Answer Choices:

A. Vm and Km will both decrease B. Vm will decrease, Km will increase C. Vm will stay the same, Km will increase D. Vm and Km will both increase

Correct Answer: C. Vm will stay the same, Km will increase

Question 60

Clinical Scenario:

A 3-year-old boy is brought to the physician for follow-up examination 5 days after sustaining a forehead laceration. Examination shows a linear, well-approximated laceration over the right temple. The wound is clean and dry with no exudate. There is a small amount of pink granulation tissue present. Microscopic examination of the wound is most likely to show which of the following?

Answer Choices:

A. Angiogenesis with type III collagen deposition B. Macrophage infiltration and fibrin clot degradation C. Capillary dilation with neutrophilic migration D. Fibroblast hyperplasia with disorganized collagen deposition

Correct Answer: A. Angiogenesis with type III collagen deposition

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Question 61

Clinical Scenario:

A 60-year-old man comes to the clinic complaining of a persistent cough for the last few months. His cough started gradually about a year ago, and it became more severe and persistent despite all his attempts to alleviate it. During the past year, he also noticed some weight loss and a decrease in his appetite. He also complains of progressive shortness of breath. He has a 40-pack-year smoking history but is a nonalcoholic. Physical examination findings are within normal limits. His chest X-ray shows a mass in the right lung. A chest CT shows a 5 cm mass with irregular borders near the lung hilum. A CT guided biopsy is planned. During the procedure, just after insertion of the needle, the patient starts to feel pain in his right shoulder. Which of the following nerves is responsible for his shoulder pain?

Answer Choices:

A. Intercostal nerves B. Phrenic nerve C. Pulmonary plexus D. Thoracic spinal nerves

Correct Answer: B. Phrenic nerve

Question 62

Clinical Scenario:

A simple experiment is performed to measure the breakdown of sucrose into glucose and fructose by a gut enzyme that catalyzes this reaction. A glucose meter is used to follow the breakdown of sucrose into glucose. When no enzyme is added to the sucrose solution, the glucose meter will have a reading of 0 mg/dL; but when the enzyme is added, the glucose meter will start to show readings indicative of glucose being formed. Which of the following diabetic pharmacological agents, when added before the addition of the gut enzyme to the sucrose solution, will maintain a reading of 0 mg/dL?

Answer Choices:

A. Glyburide B. Metformin C. Acarbose D. Exenatide

Correct Answer: C. Acarbose

Question 63

Clinical Scenario:

A 45-year-old man presents to the emergency department for sudden pain in his foot. The patient states that when he woke up, he experienced severe pain in his right great toe. The patient's wife immediately brought him to the emergency department. The patient has a past medical history of diabetes mellitus, obesity, and hypertension and is currently taking insulin, metformin, lisinopril, and ibuprofen. The patient is a current smoker and smokes 2 packs per day. He also drinks 3 glasses of whiskey every night. The patient is started on IV fluids and corticosteroids. His blood pressure, taken at the end of this visit, is 175/95 mmHg. As the patient's symptoms improve, he asks how he can avoid having these symptoms again in the future. Which of the following is the best initial intervention in preventing a future episode of this patient's condition?

Answer Choices:

A. Allopurinol B. Hydrochlorothiazide C. Lifestyle measures D. Probenecid

Correct Answer: C. Lifestyle measures

Question 64

Clinical Scenario:

A 29-year-old woman presents with low mood and tearfulness on most days for the past 4 weeks. She says that she has been struggling to cope with her life and feels that everything that is going wrong is her fault. She also says that there are nights when she cries herself to sleep as the burden of the whole day is too overwhelming for her. In the last 3 weeks, she cannot recall a day when she felt interested in going out and participating in her daily activities. She also says she doesn't seem to have much energy and feels fatigued all day. She has lost her appetite and feels that she is losing weight. Over the past month, she also reports experiencing frequent and often unbearable migraine headaches. No significant past medical history. The patient has prescribed a drug for her symptoms which is known to be cardiotoxic and may result in ECG changes. Which of the following is the mechanism of action of the drug most likely prescribed to this patient?

Answer Choices:

A. Blocks the reuptake of serotonin, increasing its concentration in the synaptic cleft B. Stimulates the release of norepinephrine and dopamine in the presynaptic terminal C. Inhibits the uptake of serotonin and norepinephrine at the presynaptic terminal D. Acts as an antagonist at the dopamine and serotonin receptors

Correct Answer: C. Inhibits the uptake of serotonin and norepinephrine at the presynaptic terminal

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Question 65

Clinical Scenario:

A 3-year-old boy is brought to the emergency room by his mother with fever and difficulty breathing after receiving the BCG vaccine. He has never had a reaction to a vaccine before. He has a history of 2 salmonella infections over the past 2 years. He was born at 35 weeks' gestation and spent one day in the neonatal intensive care unit. His parents' family histories are unremarkable. His temperature is 101°F (38.3°C), blood pressure is 80/55 mmHg, pulse is 135/min, and respirations are 24/min. On examination, he appears acutely ill. He has increased work of breathing with intercostal retractions. A petechial rash is noted on his trunk and extremities. A serological analysis in this patient would most likely reveal decreased levels of which of the following cytokines?

Answer Choices:

A. Interferon alpha B. Interferon gamma C. Interleukin 1 D. Tumor necrosis factor alpha

Correct Answer: B. Interferon gamma

Question 66

Clinical Scenario:

A 26-year-old woman with a history of asthma presents to the emergency room with persistent gnawing left lower quadrant abdominal pain. She first noticed the pain several hours ago and gets mild relief with ibuprofen. She has not traveled recently, tried any new foods or medications, or been exposed to sick contacts. She is sexually active with her boyfriend and admits that she has had multiple partners in the last year. Her temperature is 99.5°F (37.5°C), blood pressure 77/45 mmHg, pulse is 121/min, and respirations are 14/min. On exam, she appears uncomfortable and diaphoretic. She has left lower quadrant tenderness to palpation, and her genitourinary exam is normal. Her urinalysis is negative and her pregnancy test is positive. Which of the following would be the appropriate next step in management?

Answer Choices:

A. CT scan of the abdomen and pelvis B. Transvaginal ultrasound C. Administer levonorgestrel D. Exploratory laporoscopy

Correct Answer: D. Exploratory laporoscopy

Question 67

Clinical Scenario:

Five minutes after arriving in the postoperative care unit following total knee replacement under general anesthesia, a 55-year-old woman is acutely short of breath. The procedure was uncomplicated. Postoperatively, prophylactic treatment with cefazolin was begun and the patient received morphine and ketorolac for pain management. She has generalized anxiety disorder. Her only other medication is escitalopram. She has smoked one pack of cigarettes daily for 25 years. Her temperature is 37°C (98.6°F), pulse is 108/min, respirations are 26/min, and blood pressure is 95/52 mm Hg. A flow-volume loop obtained via pulmonary function testing is shown. Which of the following is the most likely underlying cause of this patient's symptoms?

Answer Choices:

A. Neuromuscular blockade B. Decreased central respiratory drive C. Bronchial hyperresponsiveness D. Type I hypersensitivity reaction

Correct Answer: D. Type I hypersensitivity reaction

Question 68

Clinical Scenario:

An 11-year-old boy is brought to the emergency room with acute abdominal pain and hematuria. Past medical history is significant for malaria. On physical examination, he has jaundice and a generalized pallor. His hemoglobin is 5 g/dL, and his peripheral blood smear reveals fragmented RBC, microspherocytes, and eccentrocytes (bite cells). Which of the following reactions catalyzed by the enzyme is most likely deficient in this patient?

Answer Choices:

A. Glucose-1-phosphate + UTP → UDP-glucose + pyrophosphate B. Glucose-6-phosphate + H_2O → glucose + Pi C. D-glucose-6-phosphate + $NADP^+$ → 6-phospho-D-glucono-1,5-lactone + NADPH + H^+ D. Glucose + ATP → Glucose-6-phosphate + ADP + H^+

Correct Answer: C. D-glucose-6-phosphate + $NADP^+$ → 6-phospho-D-glucono-1,5-lactone + NADPH + H^+

Question 69

Clinical Scenario:

An 18-year-old college student presents to the ED straight from chemistry lab where he ingested an unknown compound. He complains of a headache, and is flushed, tachypneic and tachycardic. Suspecting cyanide poisoning, you administer amyl nitrite which causes which of the following?

Answer Choices:

A. Oxidation of ferrous iron in hemoglobin to ferric iron B. A decrease in serum methemoglobin levels C. Formation of thiocyanate D. Increase in intracellular NADH/NAD+ ratio

Correct Answer: A. Oxidation of ferrous iron in hemoglobin to ferric iron

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Question 70

Clinical Scenario:

A 48-year-old man with a lengthy history of angina is brought to the emergency department after the acute onset of severe chest pain that started 40 minutes ago. Unlike previous episodes of chest pain, this one is unresponsive to nitroglycerin. His medical history is significant for hypertension, type 2 diabetes mellitus, and hyperlipidemia. His current medications include lisinopril, metformin, and simvastatin. His blood pressure is 130/80 mm Hg, heart rate is 88/min, respiratory rate is 25/min, and temperature is 36.6°C (97.8°F). An ECG shows ST-segment elevation in leads avF and V1-V3. He is administered aspirin, nasal oxygen, morphine, and warfarin; additionally, myocardial reperfusion is performed. He is discharged within 2 weeks. He comes back 3 weeks later for follow-up. Which of the following gross findings are expected to be found in the myocardium of this patient at this time?

Answer Choices:

A. Coagulation necrosis B. Red granulation tissue C. White scar tissue D. Yellow necrotic area

Correct Answer: C. White scar tissue

Question 71

Clinical Scenario:

A 69-year-old man presents for a general follow up appointment. He states that he is doing well and wants to be sure he is healthy. The patient’s past medical history is significant for type II diabetes mellitus, peripheral vascular disease, and hypertension. His current medications include metformin, glyburide, lisinopril, metoprolol and hydrochlorothiazide. His blood pressure is 130/90 mmHg and pulse is 80/min. A fasting lipid panel was performed last week demonstrating an LDL of 85 mg/dL and triglycerides of 160 mg/dL. The patient states that he has not experienced any symptoms since his last visit. The patient’s blood glucose at this visit is 100 mg/dL. Which of the following is recommended in this patient?

Answer Choices:

A. Increase lisinopril dose B. Increase metformin dose C. Begin statin therapy D. Discontinue metoprolol and start propranolol

Correct Answer: C. Begin statin therapy

Question 72

Clinical Scenario:

A 4-year-old child presents to the pediatrician with mental retardation, ataxia, and inappropriate laughter. The parents of the child decide to have the family undergo genetic testing to determine what the cause may be. The results came back and all three had no mutations that would have caused this constellation of symptoms in the child. Karyotyping was performed as well and showed no deletions, insertions, or gene translocations. Based on the symptoms, the child was diagnosed with Angelman syndrome. Which of the following genetic terms could best describe the mechanism for the disorder in the child?

Answer Choices:

A. Codominance B. Incomplete penetrance C. Uniparental disomy D. Variable expressivity

Correct Answer: C. Uniparental disomy

Question 73

Clinical Scenario:

An investigator is studying the effect of extracellular pH changes on the substrates for the citric acid cycle. Which of the following substances is required for the reaction catalyzed by the enzyme marked by the arrow in the overview of the citric acid cycle?

Answer Choices:

A. Thiamine B. Pantothenic acid C. Lipoic acid D. Niacin

Correct Answer: B. Pantothenic acid

Question 74

Clinical Scenario:

One week after delivery, a 3550-g (7-lb 13-oz) newborn has multiple episodes of bilious vomiting and abdominal distention. He passed urine 14 hours after delivery and had his first bowel movement 3 days after delivery. He was born at term to a 31-year-old woman. Pregnancy was uncomplicated and the mother received adequate prenatal care. His temperature is 37.1°C (98.8°F), pulse is 132/min, and respirations are 50/min. Examination shows a distended abdomen. Bowel sounds are hypoactive. Digital rectal examination shows a patent anus and an empty rectum. The remainder of the examination shows no abnormalities. An x-ray of the abdomen is shown. Which of the following is the underlying cause of these findings?

Answer Choices:

A. Defective migration of neural crest cells B. Disruption of blood flow to the fetal jejunum C. Mutation in the CFTR gene D. Abnormal rotation of the intestine

Correct Answer: A. Defective migration of neural crest cells

Question 75

Clinical Scenario:

In patients with chronic obstructive pulmonary disease, stimulation of muscarinic acetylcholine receptors results in an increase in mucus secretion, smooth muscle contraction and bronchoconstriction. The end result is an increase in airway resistance. Which of the following pharmacologic agents interferes directly with this pathway?

Answer Choices:

A. Epinephrine B. Theophylline C. Ipratropium D. Metoprolol

Correct Answer: C. Ipratropium

Question 76

Clinical Scenario:

A 49-year-old woman presents to her primary care physician with fatigue. She reports that she has recently been sleeping more than usual and says her “arms and legs feel like lead” for most of the day. She has gained 10 pounds over the past 3 months which she attributes to eating out at restaurants frequently, particularly French cuisine. Her past medical history is notable for social anxiety disorder. She took paroxetine and escitalopram in the past but had severe nausea and headache while taking both. She has a 10 pack-year smoking history and has several glasses of wine per day. Her temperature is 98.6°F (37°C), blood pressure is 130/65 mmHg, pulse is 78/min, and respirations are 16/min. Physical examination reveals an obese woman with a dysphoric affect. She states that her mood is sad but she does experience moments of happiness when she is with her children. The physician starts the patient on a medication to help with her symptoms. Three weeks after the initiation of the medication, the patient presents to the emergency room with a severe headache and agitation. Her temperature is 102.1°F (38.9°C), blood pressure is 180/115 mmHg, pulse is 115/min, and respirations are 24/min. Which of the following is the mechanism of action of the medication that is most likely responsible for this patient’s symptoms?

Answer Choices:

A. Inhibition of amine degradation B. Inhibition of serotonin and norepinephrine reuptake C. Partial agonism of serotonin-1A receptor D. Inhibition of serotonin reuptake

Correct Answer: A. Inhibition of amine degradation

Question 77

Clinical Scenario:

A 25-year-old G1P0000 presents to her obstetrician’s office for her first prenatal visit. She had a positive pregnancy test 6 weeks ago, and her last period was about two months ago, though at baseline her periods are irregular. Aside from some slight nausea in the mornings, she feels well. Which of the following measurements would provide the most accurate dating of this patient’s pregnancy?

Answer Choices:

A. Biparietal diameter B. Femur length C. Serum beta-hCG D. Crown-rump length

Correct Answer: D. Crown-rump length

Question 78

Clinical Scenario:

A previously healthy 8-year-old boy is brought to the physician because of increasing visual loss and deterioration of his hearing and speech over the past 2 months. During this period, he has had difficulty walking, using the stairs, and feeding himself. His teachers have noticed that he has had difficulty concentrating. His grades have worsened and his handwriting has become illegible. His maternal male cousin had similar complaints and died at the age of 6 years. Vital signs are within normal limits. Examination shows hyperpigmented skin and nails and an ataxic gait. His speech is dysarthric. Neurologic examination shows spasticity and decreased muscle strength in all extremities. Deep tendon reflexes are 4+ bilaterally. Plantar reflex shows an extensor response bilaterally. Sensation is decreased in the lower extremities. Fundoscopy shows optic atrophy. There is sensorineural hearing loss bilaterally. Which of the following is the most likely cause of this patient's symptoms?

Answer Choices:

A. β-Glucocerebrosidase deficiency B. ATP-binding cassette transporter dysfunction C. Arylsulfatase A deficiency D. α-Galactosidase A deficiency

Correct Answer: B. ATP-binding cassette transporter dysfunction

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Question 79

Clinical Scenario:

An investigator is studying the incidence of sickle cell trait in African American infants. To identify the trait, polymerase chain reaction testing is performed on venous blood samples obtained from the infants. Which of the following is required for this laboratory technique?

Answer Choices:

A. RNA-dependent DNA polymerase B. Ligation of Okazaki fragments C. Initial sequence of the 3' end of a DNA strand D. Complete genome DNA sequence

Correct Answer: C. Initial sequence of the 3' end of a DNA strand

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Question 80

Clinical Scenario:

A 75-year-old man presents to the physician with a complaint of persistent back pain. The patient states that the pain has been constant and occurs throughout the day. He says that he has also been experiencing greater fatigue when carrying out his daily activities. On review of systems, the patient notes that he lost more than 10 pounds in the past month despite maintaining his usual diet and exercising less often due to his fatigue. Physical exam is notable for a systolic murmur at the right sternal border, mild crackles at the bases of both lungs, and tenderness to palpation of his lumbar spine. Laboratory values are below:

Serum: Na+: 141 mEq/L Cl-: 101 mEq/L K+: 4.2 mEq/L HCO3-: 23 mEq/L BUN: 20 mg/dL Glucose: 101 mg/dL Creatinine: 1.6 mg/dL Ca2+: 12.8 mg/dL

A peripheral blood smear is ordered for the patient’s work-up. Which of the following would be the most likely finding on peripheral blood smear?

Answer Choices:

A. Atypical lymphocytes B. Rouleaux formation C. Schistocytes D. Target cells

Correct Answer: B. Rouleaux formation

Question 81

Clinical Scenario:

A 17-year-old boy is brought to the physician because of progressive right knee pain for the past 3 months. He reports that the pain is worse at night and while doing sports at school. He has not had any trauma to the knee or any previous problems with his joints. His vital signs are within normal limits. Examination of the right knee shows mild swelling and tenderness without warmth or erythema; the range of motion is limited. He walks with an antalgic gait. Laboratory studies show an alkaline phosphatase of 180 U/L and an erythrocyte sedimentation rate of 80 mm/h. An x-ray of the right knee is shown. Which of the following is the most likely diagnosis?

Answer Choices:

A. Ewing sarcoma B. Chordoma C. Chondrosarcoma D. Osteosarcoma "

Correct Answer: D. Osteosarcoma "

Question 82

Clinical Scenario:

A 44-year-old man presents to the emergency department with weakness. He states that he has felt progressively more weak over the past month. He endorses decreased libido, weight gain, and headaches. His temperature is 97.0°F (36.1°C), blood pressure is 177/108 mmHg, pulse is 80/min, respirations are 17/min, and oxygen saturation is 98% on room air. Physical exam is notable for an obese man who appears fatigued. He has abdominal striae, atrophied arms, and limbs with minimal muscle tone. His ECG is notable for a small upward deflection right after the T wave. A fingerstick blood glucose is 225 mg/dL. The patient is treated appropriately and states that he feels much better several hours later. Which of the following treatments could prevent this patient from presenting again with a similar chief complaint?

Answer Choices:

A. Eplerenone B. Hydrochlorothiazide C. Insulin D. Torsemide

Correct Answer: A. Eplerenone

Question 83

Clinical Scenario:

A 63-year-old woman comes to the office because of a 2-year history of upper and lower extremity weakness and neck pain that is worse with sneezing. She has had difficulty swallowing and speaking for the past 8 months. Musculoskeletal examination shows spasticity and decreased muscle strength in all extremities. There is bilateral atrophy of the trapezius and sternocleidomastoid muscles. Neurologic examination shows an ataxic gait and dysarthria. Deep tendon reflexes are 4+ bilaterally. Babinski sign is positive. Sensation is decreased below the C5 dermatome bilaterally. An MRI of the neck and base of the skull is shown. Which of the following is the most likely cause of this patient's symptoms?

Answer Choices:

A. Foramen magnum meningioma B. Cerebral glioblastoma multiforme C. Amytrophic lateral sclerosis D. Syringomyelia "

Correct Answer: A. Foramen magnum meningioma

Question 84

Clinical Scenario:

An investigator is studying the effects of influenza virus on human lung tissue. Biopsy specimens of lung parenchyma are obtained from patients recovering from influenza pneumonia and healthy control subjects. Compared to the lung tissue from the healthy control subjects, the lung tissue from the affected patients is most likely to show which of the following findings on histopathologic examination?

Answer Choices:

A. Decreased alveolar macrophages B. Decreased interstitial fibroblasts C. Increased type II pneumocytes D. Increased goblet cells

Correct Answer: C. Increased type II pneumocytes

Question 85

Clinical Scenario:

A 32-year-old G0P0 African American woman presents to the physician with complaints of heavy menstrual bleeding as well as menstrual bleeding in between her periods. She also reports feeling fatigued and having bizarre cravings for ice and chalk. Despite heavy bleeding, she does not report any pain with menstruation. Physical examination is notable for an enlarged, asymmetrical, firm uterus with multiple palpable, non-tender masses. Biopsy confirms the diagnosis of a benign condition. Which of the following histological characteristics would most likely be seen on biopsy in this patient?

Answer Choices:

A. Clustered pleomorphic, hyperchromatic smooth muscle cells with extensive mitosis B. Laminated, concentric spherules with dystrophic calcification C. Presence of endometrial glands and stroma in the myometrium D. Whorled pattern of smooth muscle bundles with well-defined borders

Correct Answer: D. Whorled pattern of smooth muscle bundles with well-defined borders

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Question 86

Clinical Scenario:

A 38-year-old woman presents to her primary care physician for evaluation of 3 months of increasing fatigue. She states that she feels normal in the morning, but that her fatigue gets worse throughout the day. Specifically, she says that her head drops when trying to perform overhead tasks. She also says that she experiences double vision when watching television or reading a book. On physical exam, there is right-sided ptosis after sustaining upward gaze for a 2 minutes. Which of the following treatments may be effective in treating this patient's diagnosis?

Answer Choices:

A. Antitoxin B. Chemotherapy C. Thymectomy D. Vaccination

Correct Answer: C. Thymectomy

Question 87

Clinical Scenario:

A 4-day-old male newborn is brought to the physician for a well-child examination. His mother is concerned that he is losing weight. He was born at 40 weeks' gestation and weighed 2980g (6-lb 9-oz); he currently weighs 2830g (6-lb 4-oz). Pregnancy was uncomplicated. He passed stool and urine 8 and 10 hours after delivery. He has been exclusively breast fed since birth and feeds 11–12 times daily. His mother says she changes 5–6 heavy diapers daily. Examination shows an open and firm anterior fontanelle. Mucous membranes are moist. Capillary refill time is less than 2 seconds. Cardiopulmonary examination shows no abnormalities. Which of the following is the most appropriate next best step in management?

Answer Choices:

A. Serum creatinine and urea nitrogen B. Add rice based cereal C. Add cow milk based formula D. Continue breastfeeding "

Correct Answer: D. Continue breastfeeding "

Question 88

Clinical Scenario:

A 78-year-old woman is accompanied by her family for a routine visit to her primary care provider. The family states that 5 months prior, the patient had a stroke and is currently undergoing physical therapy. Today, her temperature is 98.2°F (36.8°C), blood pressure is 112/72 mmHg, pulse is 64/min, and respirations are 12/min. On exam, she is alert and oriented with no deficits in speech. Additionally, her strength and sensation are symmetric and preserved bilaterally. However, on further neurologic testing, she appears to have some difficulty with balance and a propensity to fall to her right side. Which of the following deficits does the patient also likely have?

Answer Choices:

A. Contralateral eye deviation B. Hemiballismus C. Intention tremor D. Truncal ataxia

Correct Answer: C. Intention tremor

Question 89

Clinical Scenario:

A 65-year-old male with a history of COPD presents to the emergency department with dyspnea, productive cough, and a fever of 40.0°C (104.0°F) for the past 2 days. His respiratory rate is 20/min, blood pressure is 125/85 mm Hg, and heart rate is 95/min. A chest X-ray is obtained and shows a right lower lobe infiltrate. Sputum cultures are pending and he is started on antibiotics. The patient has not received any vaccinations in the last 20 years. The physician discusses with him the importance of getting a vaccine that can produce immunity via which of the following mechanisms?

Answer Choices:

A. T cell-dependent B cell response B. Natural killer cell response C. Mast cell degranulation response D. No need to vaccinate, as the patient has already had a pneumonia vaccine

Correct Answer: A. T cell-dependent B cell response

Question 90

Clinical Scenario:

A 42-year-old woman is in the hospital recovering from a cholecystectomy performed 3 days ago that was complicated by cholangitis. She is being treated with IV piperacillin-tazobactam. She calls the nurse to her room because she says that her heart is racing. She also demands that someone come in to clean the pile of garbage off of the floor because it is attracting flies. Her pulse is 112/min, respiratory rate is 20/min, temperature is 38.0°C (100.4°F), and blood pressure is 150/90 mm Hg. On physical examination, the patient appears sweaty, distressed, and unable to remain still. She is oriented to person, but not place or time. Palpation of the abdomen shows no tenderness, rebound, or guarding. Which of the following is the most likely diagnosis in this patient?

Answer Choices:

A. Acute cholangitis B. Alcoholic hallucinosis C. Delirium tremens D. Hepatic encephalopathy

Correct Answer: C. Delirium tremens

Question 91

Clinical Scenario:

A 37-year-old G1P1001 presents for her 6-week postpartum visit after delivering a male infant by spontaneous vaginal delivery at 41 weeks and 5 days gestation. She notes that five days ago, her right breast began to hurt, and the skin near her nipple turned red. She also states that she has felt feverish and generally achy for 2 days but thought she was just sleep deprived. The patient's son has been having difficulty latching for the last 2 weeks and has begun receiving formula in addition to breast milk, though the patient wishes to continue breastfeeding. She is generally healthy with no past medical history but has smoked half a pack per day for the last 15 years. Her mother died from breast cancer at the age of 62, and her father has hypertension and coronary artery disease. At this visit, her temperature is 100.6° F (38.1° C), blood pressure is 116/73 mmHg, pulse is 80/min, and respirations are 14/min. She appears tired and has a slightly flat affect. Examination reveals a 4x4 cm area of erythema on the lateral aspect near the nipple on the right breast. In the center of this area, there is a fluctuant, tender mass that measures 2x2 cm. The overlying skin is intact. The remainder of her exam is unremarkable. Which of the following is the best next step in management?

Answer Choices:

A. Mammogram B. Incision and drainage C. Needle aspiration and oral dicloxacillin D. Cessation of smoking

Correct Answer: C. Needle aspiration and oral dicloxacillin

Question 92

Clinical Scenario:

A 3-year-old boy is brought to the physician for a follow-up examination. He has suffered from seizures since the age of 8 months. His mother has noticed he often has unprovoked bouts of laughter and loves playing with water. She describes him as having a happy, excitable demeanor. He can stand without support but cannot walk. His responses are rarely verbal, and when they are, he uses single words only. His only medication is sodium valproate. He is at the 2nd percentile for head circumference, 30th percentile for height, and 60th percentile for weight. Examination shows a wide-based stance and mandibular prognathism. Tongue thrusting and difficulty standing is present. Muscle tone is increased in all extremities. Deep tendon reflexes are 4+ bilaterally. Which of the following is the mechanism most likely to explain these findings?

Answer Choices:

A. Microdeletion of maternal 15q11-q13 B. Microdeletion of paternal 15q11-q13 C. MECP2 gene mutation D. Microdeletion of 22q11.2

Correct Answer: A. Microdeletion of maternal 15q11-q13

Question 93

Clinical Scenario:

A 49-year-old male presents to the emergency room with dyspnea and pulmonary edema. He reports that he has been smoking 2 packs a day for the past 25 years and has difficulty breathing during any sustained physical activity. His blood pressure is normal, and he reports a history of COPD. An echocardiogram was ordered as part of a cardiac workup. Which of the following would be the most likely finding?

Answer Choices:

A. Aortic stenosis B. Mitral valve insufficiency C. Coronary sinus dilation D. Tricuspid valve stenosis

Correct Answer: C. Coronary sinus dilation

Question 94

Clinical Scenario:

A 27-year-old male arrives to your walk-in clinic complaining of neck pain. He reports that the discomfort began two hours ago, and now he feels like he can’t move his neck. He also thinks he is having hot flashes, but he denies dyspnea or trouble swallowing. The patient’s temperature is 99°F (37.2°C), blood pressure is 124/76 mmHg, pulse is 112/min, and respirations are 14/min with an oxygen saturation of 99% O_2 on room air. You perform a physical exam of the patient's neck, and you note that his neck is rigid and flexed to the left. You are unable to passively flex or rotate the patient's neck to the right. There is no airway compromise. The patient's past medical history is significant for asthma, and he was also recently diagnosed with schizophrenia. The patient denies current auditory or visual hallucinations. He appears anxious, but his speech is organized and appropriate. Which of the following is the best initial step in management?

Answer Choices:

A. Change medication to clozapine B. Dantrolene C. Diphenhydramine D. Propranolol

Correct Answer: C. Diphenhydramine

Question 95

Clinical Scenario:

A 7-year-old girl is brought to the physician with complaints of erythema and rashes over the bridge of her nose and on her forehead for the past 6 months. She also has vesiculobullous and erythematous scaly crusted lesions on the scalp and around the perioral areas. Her parents report a history of worsening symptoms during exposure to sunlight, along with a history of joint pain and oral ulcers. Her temperature is 38.6°C (101.4°F), pulse is 88/min, and respirations are 20/min. On physical examination, pallor and cervical lymphadenopathy are present. On cutaneous examination, diffuse hair loss and hyperpigmented scaly lesions are present. Her laboratory studies show: Hemoglobin 7.9 mg/dL Total leukocyte count 6,300/mm3 Platelet count 167,000/mm3 Erythrocyte sedimentation rate 30 mm/h ANA titer 1:520 (positive) Which of the following most likely explains the mechanism of this condition?

Answer Choices:

A. Type I hypersensitivity B. Type II hypersensitivity C. Type III hypersensitivity D. Type IV hypersensitivity

Correct Answer: C. Type III hypersensitivity

Question 96

Clinical Scenario:

A 21-year-old college student comes to the physician for intermittent palpitations. She does not have chest pain or shortness of breath. The symptoms started 2 days ago, on the night after she came back to her dormitory after a 4-hour-long bus trip from home. A day ago, she went to a party with friends. The palpitations have gotten worse since then and occur more frequently. The patient has smoked 5 cigarettes daily for the past 3 years. She drinks 4–6 alcoholic beverages with friends once or twice a week and occasionally uses marijuana. She is sexually active with her boyfriend and takes oral contraceptive pills. She does not appear distressed. Her pulse is 100/min and irregular, blood pressure is 140/85 mm Hg, and respirations are at 25/min. Physical examination shows a fine tremor in both hands, warm extremities, and swollen lower legs. The lungs are clear to auscultation. An ECG is shown below. Which of the following is the most appropriate next step in management?

Answer Choices:

A. Measure TSH levels B. Observe and wait C. Measure D-Dimer levels D. Send urine toxicology

Correct Answer: A. Measure TSH levels

Question 97

Clinical Scenario:

A 45-year-old male presents to the emergency room complaining of severe diarrhea. He recently returned from a business trip to Bangladesh. Since returning, he has experienced several loose bloody stools per day that are accompanied by abdominal cramping and occasional nausea and vomiting. His temperature is 101.7°F (38.7°C), blood pressure is 100/60 mmHg, pulse is 120/min, and respirations are 20/min. On examination, he demonstrates mild tenderness to palpation throughout his abdomen, delayed capillary refill, and dry mucus membranes. Results from a stool sample and subsequent stool culture are pending. What is the mechanism of action of the toxin elaborated by the pathogen responsible for this patient's current condition?

Answer Choices:

A. ADP-ribosylation of elongation factor 2 B. Stimulation of guanylyl cyclase C. ADP-ribosylation of a G protein D. Inhibition of 60S ribosomal subunit

Correct Answer: D. Inhibition of 60S ribosomal subunit

Question 98

Clinical Scenario:

A 23-year-old gravida-1-para-1 (G1P1) presents to the emergency department with fever, malaise, nausea, and abdominal pain. She says her symptoms started 2 days ago with a fever and nausea, which have progressively worsened. 2 hours ago, she started having severe lower abdominal pain that is diffusely localized. Her past medical history is unremarkable. Her last menstrual period was 3 weeks ago. She has had 3 sexual partners in the past month and uses oral contraception. The vital signs include temperature 38.8°C (102.0°F) and blood pressure 120/75 mm Hg. On physical examination, the lower abdomen is severely tender to palpation with guarding. Uterine and adnexal tenderness is also elicited. A urine pregnancy test is negative. On speculum examination, the cervix is inflamed with positive cervical motion tenderness and the presence of a scant yellow-white purulent discharge. Which of the following is the most likely diagnosis in this patient?

Answer Choices:

A. Vaginitis B. Cervicitis C. Pelvic inflammatory disease D. Urinary tract infection

Correct Answer: C. Pelvic inflammatory disease

Question 99

Clinical Scenario:

A 40-year-old man is physically and verbally abusive towards his wife and two children. When he was a child, he and his mother were similarly abused by his father. Which of the following psychological defense mechanisms is this man demonstrating?

Answer Choices:

A. Identification B. Distortion C. Projection D. Splitting

Correct Answer: A. Identification

Question 100

Clinical Scenario:

A 25-year-old man visits a local clinic while volunteering abroad to rebuild homes after a natural disaster. He reports that he has been experiencing an intermittent rash on his feet for several weeks that is associated with occasional itching and burning. He states that he has been working in wet conditions in work boots and often does not get a chance to remove them until just before going to bed. On physical exam, there is diffuse erythema and maceration of the webspaces between his toes. He starts taking a medication. Two days later, he experiences severe nausea and vomiting after drinking alcohol. Which of the following is the mechanism of action of the drug most likely prescribed in this case?

Answer Choices:

A. Cell arrest at metaphase B. Disruption of fungal cell membrane C. Inhibition of cell wall synthesis D. Inhibition of DNA synthesis

Correct Answer: A. Cell arrest at metaphase

Question 101

Clinical Scenario:

A 32-year-old man is brought to the emergency department because he was found stumbling in the street heedless of oncoming traffic. On arrival, he is found to be sluggish and has slow and sometimes incoherent speech. He is also drowsy and falls asleep several times during questioning. Chart review shows that he has previously been admitted after getting a severe cut during a bar fight. Otherwise, he is known to be intermittently homeless and has poorly managed diabetes. Serum testing reveals the presence of a substance that increases the duration of opening for an important channel. Which of the following symptoms may be seen if the most likely substance in this patient is abruptly discontinued?

Answer Choices:

A. Cardiovascular collapse B. Flashbacks C. Insomnia D. Piloerection

Correct Answer: A. Cardiovascular collapse

Question 102

Clinical Scenario:

A 38-year-old man comes to the physician because of a 2-year-history of cough and progressively worsening breathlessness. He has smoked 1 pack of cigarettes daily for the past 10 years. Physical examination shows contraction of the anterior scalene and sternocleidomastoid muscles during inspiration. An x-ray of the chest shows flattening of the diaphragm and increased radiolucency in the lower lung fields. Further analysis shows increased activity of an isoform of elastase that is normally inhibited by alpha-1-antitrypsin. The cells that produce this isoform of elastase were most likely stimulated to enter the site of inflammation by which of the following substances?

Answer Choices:

A. Lactoferrin B. Interferon gamma C. Leukotriene B4 D. Thromboxane A2

Correct Answer: C. Leukotriene B4

Question 103

Clinical Scenario:

A 35-year-old woman with irritable bowel syndrome comes to the physician because of increased diarrhea. She has not had any fever, bloody stools, nausea, or vomiting. The increase in stool frequency began when she started a new job. She is started on loperamide, and her symptoms improve. Which of the following is the primary mechanism of action of this drug?

Answer Choices:

A. μ-opioid receptor agonism B. 5-HT3 receptor antagonism C. Acetylcholine receptor antagonism D. Physical protection of stomach mucosa

Correct Answer: A. μ-opioid receptor agonism

Question 104

Clinical Scenario:

A 4-year-old girl is brought to the emergency department by her parents with a sudden onset of breathlessness. She has been having similar episodes over the past few months with a progressive increase in frequency over the past week. They have noticed that the difficulty in breathing is more prominent during the day when she plays in the garden with her siblings. She gets better once she comes indoors. During the episodes, she complains of an inability to breathe and her parents say that she is gasping for breath. Sometimes they hear a noisy wheeze while she breathes. The breathlessness does not disrupt her sleep. On examination, she seems to be in distress with noticeable intercostal retractions. Auscultation reveals a slight expiratory wheeze. According to her history and physical findings, which of the following mechanisms is most likely responsible for this child’s difficulty in breathing?

Answer Choices:

A. Destruction of the elastic layers of bronchial walls leading to abnormal dilation B. Defective chloride channel function leading to mucus plugging C. Inflammation leading to permanent dilation and destruction of alveoli D. Airway hyperreactivity to external allergens causing intermittent airway obstruction

Correct Answer: D. Airway hyperreactivity to external allergens causing intermittent airway obstruction

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Question 105

Clinical Scenario:

A 39-year-old female presents to her gynecologist complaining of a breast lump. Two weeks ago, while performing a breast self-examination she noticed a small firm nodule in her left breast. She is otherwise healthy and takes no medications. Her family history is notable for a history of breast cancer in her mother and maternal aunt. On physical examination, there is a firm immobile nodular mass in the superolateral quadrant of her left breast. A mammogram of her left breast is shown. Genetic analysis reveals a mutation on chromosome 17. This patient is at increased risk for which of the following conditions?

Answer Choices:

A. Serous cystadenocarcinoma B. Granulosa-theca cell tumor C. Uterine leiomyosarcoma D. Transitional cell carcinoma

Correct Answer: A. Serous cystadenocarcinoma

Question 106

Clinical Scenario:

A 54-year-old man comes to the physician for the evaluation of difficulty swallowing of both solids and liquids for 1 month. During the past 5 months, he has also had increased weakness of his hands and legs. He sails regularly and is unable to hold the ropes as tightly as before. Ten years ago, he was involved in a motor vehicle collision. Examination shows atrophy of the tongue. Muscle strength is decreased in the right upper and lower extremities. There is muscle stiffness in the left lower extremity. Deep tendon reflexes are 1+ in the right upper and lower extremities, 3+ in the left upper extremity, and 4+ in the left lower extremity. Plantar reflex shows an extensor response on the left foot. Sensation to light touch, pinprick, and vibration is intact. Which of the following is the most likely diagnosis?

Answer Choices:

A. Amyotrophic lateral sclerosis B. Inclusion-body myositis C. Subacute combined degeneration of spinal cord D. Cervical spondylosis with myelopathy "

Correct Answer: A. Amyotrophic lateral sclerosis

Question 107

Clinical Scenario:

A 28-year-old woman with a history of migraines presents to your office due to sudden loss of vision in her left eye and difficulty speaking. Two weeks ago she experienced muscle aches, fever, and cough. Her muscle aches are improving but she continues to have a cough. She also feels as though she has been more tired than usual. She had a similar episode of vision loss 2 years ago and had an MRI at that time. She has a family history of migraines and takes propranolol daily. On swinging light test there is decreased constriction of the left pupil relative to the right pupil. You repeat the MRI and note enhancing lesions in the left optic nerve. Which of the following is used to prevent progression of this condition?

Answer Choices:

A. Dexamethasone B. Methotrexate C. Natalizumab D. Adalimumab

Correct Answer: C. Natalizumab

Question 108

Clinical Scenario:

A 63-year-old man presents to his family physician with limited movement in his left shoulder that has progressed gradually over the past 6 years. He previously had pain when moving his shoulder, but the pain subsided a year ago and now he experiences the inability to fully flex, abduct, and rotate his left arm. He had an injury to his left shoulder 10 years ago when he fell onto his arms and 'stretched ligaments'. He did not seek medical care and managed the pain with NSAIDs and rest. He has diabetes mellitus that is well controlled with Metformin. His blood pressure is 130/80 mm Hg, the heart rate is 81/min, the respiratory rate is 15/min, and the temperature is 36.6°C (97.9°F). Physical examination reveals limitations of both active and passive abduction and external rotation in the left arm. The range of motion in the right glenohumeral joint is normal. The muscles of the left shoulder look less bulky than those of the right shoulder. There is no change in shoulder muscle power bilaterally. The reflexes and sensation on the upper extremities are normal. Which of the following is the next best step for this patient?

Answer Choices:

A. NSAID prescription for 1–2 weeks B. Physical therapy C. Corticosteroid injections D. Arthroscopic capsular release

Correct Answer: B. Physical therapy

Question 109

Clinical Scenario:

A previously healthy 30-year-old woman comes to the physician because of a 3-month history of progressive shortness of breath and nonproductive cough. She also complains of constipation and fatigue during the same time period. She has not traveled recently or been exposed to any sick contacts. Physical examination shows injected conjunctivae and tender, erythematous nodules on both shins. The lungs are clear to auscultation. An x-ray of the chest is shown. Which of the following additional findings is most likely in this patient?

Answer Choices:

A. Positive interferon-gamma release assay B. Low serum angiotensin-converting enzyme levels C. Low serum CD4+ T-cell count D. Positive anti-dsDNA antibody testing

Correct Answer: C. Low serum CD4+ T-cell count

Question 110

Clinical Scenario:

A 51-year-old woman comes to the physician because of progressively worsening lower back pain. The pain radiates down the right leg to the lateral side of the foot. She has had no trauma, urinary incontinence, or fever.

An MRI of the lumbar spine shows disc degeneration and herniation at the level of L5–S1. Which of the following is the most likely finding on physical examination?

Answer Choices:

A. Difficulty walking on heels B. Exaggerated patellar tendon reflex C. Weak achilles tendon reflex D. Diminished sensation of the anterior lateral thigh "

Correct Answer: C. Weak achilles tendon reflex

Question 111

Clinical Scenario:

A previously healthy 14-year-old girl is brought to the emergency department by her mother because of abdominal pain, nausea, and vomiting for 6 hours. Over the past 6 weeks, she has also had increased frequency of urination, and she has been drinking more water than usual. She has lost 6 kg (13 lb) over the same time period despite having a good appetite. Her temperature is 37.1°C (98.8°F), pulse is 125/min, respirations are 32/min, and blood pressure is 94/58 mm Hg. She appears lethargic. Physical examination shows deep and labored breathing and dry mucous membranes. The abdomen is soft, and there is diffuse tenderness to palpation with no guarding or rebound. Urine dipstick is positive for ketones and glucose. Further evaluation is most likely to show which of the following findings?

Answer Choices:

A. Increased arterial pCO_2 B. Increased arterial blood pH C. Serum glucose concentration > 800 mg/dL D. Decreased total body potassium

Correct Answer: D. Decreased total body potassium

Question 112

Clinical Scenario:

A 23-year-old woman is brought to the emergency department by her friend because of strange behavior. Two hours ago, she was at a night club where she got involved in a fight with the bartender. Her friend says that she was smoking a cigarette before she became irritable and combative. She repeatedly asked “Why are you pouring blood in my drink?” before hitting the bartender. She has no history of psychiatric illness. Her temperature is 38°C (100.4°F), pulse is 100/min, respirations are 19/min, and blood pressure is 158/95 mm Hg. Examination shows muscle rigidity. She has a reduced degree of facial expression. She has no recollection of her confrontation with the bartender. Which of the following is the most likely primary mechanism responsible for this patient's symptoms?

Answer Choices:

A. Stimulation of cannabinoid receptors B. Stimulation of 5HT2A and dopamine D2 receptors C. Inhibition of norepinephrine, serotonin, and dopamine reuptake D. Inhibition of NMDA receptors

Correct Answer: D. Inhibition of NMDA receptors

Question 113

Clinical Scenario:

A healthy 33-year-old gravida 1, para 0, at 15 weeks' gestation comes to the genetic counselor for a follow-up visit. Her uncle had recurrent pulmonary infections, chronic diarrhea, and infertility, and died at the age of 28 years. She does not smoke or drink alcohol. The results of an amniotic karyotype analysis show a deletion of Phe508 on chromosome 7. This patient's fetus is at greatest risk for developing which of the following complications?

Answer Choices:

A. Congenital megacolon B. Cardiac defects C. Meconium ileus D. Neural tube defects

Correct Answer: C. Meconium ileus

Question 114

Clinical Scenario:

A group of researchers studying the relationship between major depressive disorder and unprovoked seizures identified 36 patients via chart review who had been rehospitalized for unprovoked seizures following discharge from an inpatient psychiatric unit and 105 patients recently discharged from the same unit who did not experience unprovoked seizures. The results of the study show: Unprovoked seizure No seizure Major depressive disorder 20 35 No major depressive disorder 16 70 Based on this information, which of the following is the most appropriate measure of association between history of major depressive disorder (MDD) and unprovoked seizures?"

Answer Choices:

A. 0.36 B. 1.95 C. 2.5 D. 0.17

Correct Answer: C. 2.5

Question 115

Clinical Scenario:

A 45-year-old primigravida woman at 13-weeks' gestation is scheduled for a prenatal evaluation. This is her first appointment, though she has known she is pregnant for several weeks. A quad screening is performed with the mother's blood and reveals the following: AFP (alpha-fetoprotein) Decreased hCG (human chorionic gonadotropin) Elevated Estriol Decreased Inhibin Elevated Ultrasound evaluation of the fetus reveals increased nuchal translucency. Which mechanism of the following mechanisms is most likely to have caused the fetus’s condition?

Answer Choices:

A. Robertsonian translocation B. Nondisjunction C. Nucleotide excision repair defect D. Mosaicism

Correct Answer: B. Nondisjunction

Question 116

Clinical Scenario:

An 8-year-old African American girl is brought to the clinic by her mother for her regular blood exchange. They come in every 2–3 months for the procedure. The child is in good health with no symptoms. Her last trip to the emergency department was 6 months ago due to bone pain. She was treated with morphine and oxygen and a blood transfusion. She takes hydroxyurea and a multivitamin with iron every day. She has an uncle that also has to get blood exchanges. Today, her heart rate is 90/min, respiratory rate is 17/min, blood pressure is 110/65 mm Hg, and temperature is 37.0°C (98.6°F). She calmly waits for the machine to be set up and catheters inserted into both of her arms. She watches a movie as her blood is slowly replaced with 6 L of red blood cells. Based on this history, which of the following mechanisms most likely explains this patient’s condition?

Answer Choices:

A. Amino acid deletion B. Amino acid substitution C. Trinucleotide repeat D. Nonsense mutation

Correct Answer: B. Amino acid substitution

Question 117

Clinical Scenario:

A 78-year-old woman comes to her family physician for an annual health maintenance examination. Her husband, who worked as an art collector and curator, recently passed away. To express her gratitude for the longstanding medical care of her husband, she offers the physician and his staff a framed painting from her husband's art collection. Which of the following is the most appropriate reaction by the physician?

Answer Choices:

A. Accept the gift to maintain a positive patient-physician relationship but decline any further gifts. B. Politely decline and explain that he cannot accept valuable gifts from his patients. C. Accept the gift and donate the painting to a local museum. D. Accept the gift and assure the patient that he will take good care of her.

Correct Answer: B. Politely decline and explain that he cannot accept valuable gifts from his patients.

Question 118

Clinical Scenario:

A 27-year-old female in her 20th week of pregnancy presents for a routine fetal ultrasound screening. An abnormality of the right fetal kidney is detected. It is determined that the right ureteropelvic junction has failed to recanalize. Which of the following findings is most likely to be seen on fetal ultrasound:

Answer Choices:

A. Unilateral hydronephrosis B. Renal cysts C. Pelvic kidney D. Duplicated ureter

Correct Answer: A. Unilateral hydronephrosis

Question 119

Clinical Scenario:

A 46-year-old woman presents to her primary care physician for her annual examination. At her prior exam one year earlier, she had a Pap smear which was within normal limits. Which of the following health screenings is recommended for this patient?

Answer Choices:

A. Blood glucose and/or HbA1c screening B. Yearly Pap smear C. Bone mineral density screening D. Colorectal screening

Correct Answer: A. Blood glucose and/or HbA1c screening

Question 120

Clinical Scenario:

A 59-year-old man comes to the physician because of urinary frequency and perineal pain for the past 3 days. During this time, he has also had pain with defecation. He is sexually active with his wife only. His temperature is 39.1°C (102.3°F). His penis and scrotum appear normal. Digital rectal examination shows a swollen, exquisitely tender prostate. His leukocyte count is 13,400/mm3. A urine culture obtained prior to initiating treatment is most likely to show which of the following?

Answer Choices:

A. Gram-negative, lactose-fermenting rods in pink colonies B. Gram-negative, oxidase-positive rods in green colonies C. Gram-negative, encapsulated rods in mucoid colonies D. Gram-negative, aerobic, intracellular diplococci

Correct Answer: A. Gram-negative, lactose-fermenting rods in pink colonies

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Question 121

Clinical Scenario:

A 68-year-old man is brought to the emergency department because of right-sided weakness for 2 hours. He has hypertension, dyslipidemia, and type 2 diabetes. Current medications include hydrochlorothiazide, metoprolol, amlodipine, pravastatin, and metformin. His pulse is 87/min and blood pressure is 164/98 mm Hg. Neurological examination shows right-sided weakness, facial droop, and hyperreflexia. Sensation is intact. Which of the following is the most likely cause of these findings?

Answer Choices:

A. Rupture of an intracranial aneurysm B. Lipohyalinosis of penetrating vessels C. Stenosis of the internal carotid artery D. Embolism from the left atrium

Correct Answer: B. Lipohyalinosis of penetrating vessels

Question 122

Clinical Scenario:

An investigator is studying a hereditary defect in the mitochondrial enzyme succinyl-CoA synthetase. In addition to succinate, the reaction catalyzed by this enzyme produces a molecule that is utilized as an energy source for protein translation. This molecule is also required for which of the following conversion reactions?

Answer Choices:

A. Acetaldehyde to acetate B. Fructose-6-phosphate to fructose-1,6-bisphosphate C. Glucose-6-phosphate to 6-phosphogluconolactone D. Oxaloacetate to phosphoenolpyruvate

Correct Answer: D. Oxaloacetate to phosphoenolpyruvate

Question 123

Clinical Scenario:

Positron emission tomography is conducted and indicates a malignant nodule. Bronchoscopy with transbronchial biopsy is performed and a specimen sample of the nodule is sent for frozen section analysis. The tissue sample is most likely to show which of the following pathohistological findings?

Answer Choices:

A. Large cell carcinoma B. Carcinoid tumor C. Squamous cell carcinoma D. Metastasis of colorectal cancer

Correct Answer: C. Squamous cell carcinoma

Question 124

Clinical Scenario:

A 62-year-old woman comes to the physician for a follow-up examination after a recent change in her medication regimen. She reports that she feels well. She has type 2 diabetes mellitus, hyperlipidemia, hypertension, essential tremor, and chronic back pain. Current medications are metformin, glyburide, propranolol, simvastatin, ramipril, amitriptyline, and ibuprofen. Fingerstick blood glucose concentration is 47 mg/dL. Serum studies confirm this value. Which of the following pharmacologic mechanisms is most likely responsible for the absence of symptoms in this patient?

Answer Choices:

A. Inhibition of norepinephrine and serotonine reuptake B. Antagonism at β2-adrenergic receptors C. Inhibition of angiotensin-converting enzyme D. Inhibition of HMG-CoA reductase

Correct Answer: B. Antagonism at β2-adrenergic receptors

Question 125

Clinical Scenario:

A 65-year-old man is brought to his primary care provider by his concerned wife. She reports he has had this "thing" on his eye for years and refuses to seek care. He denies any pain or discharge from the affected eye. A picture of his eye is shown below. Given the diagnosis, what are you most likely to discover when taking this patient's history?

Answer Choices:

A. He experienced shingles three years ago, with a positive Hutchinson's sign B. He suffered from recurrent conjunctivitis in his youth C. He grew up in Ecuador, where he worked outdoors as a farmer for 30 years D. He suffered a burn to his eye while cleaning his bathroom with bleach 5 years earlier

Correct Answer: C. He grew up in Ecuador, where he worked outdoors as a farmer for 30 years

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Question 126

Clinical Scenario:

A 15-year-old boy is brought to the physician for a well-child visit. His parents are concerned that he has not had his growth spurt yet. As a child, he was consistently in the 60th percentile for height; now he is in the 25th percentile. His classmates make fun of his height and high-pitched voice. His parents are also concerned that he does not maintain good hygiene. He frequently forgets to shower and does not seem aware of his body odor. As an infant, he had bilateral orchidopexy for cryptorchidism and a cleft palate repair. He is otherwise healthy. Vital signs are within normal limits. On physical exam, axillary and pubic hair is sparse. Genitals are Tanner stage 1 and the testicles are 2 mL bilaterally. Which of the following is the most likely diagnosis?

Answer Choices:

A. Hyperprolactinemia B. Hypothyroidism C. Primary hypogonadism D. Kallmann syndrome

Correct Answer: D. Kallmann syndrome

Question 127

Clinical Scenario:

A 21-year-old woman has frequent sexual fantasies about female coworkers. When she is with her friends in public, she never misses an opportunity to make derogatory comments about same-sex couples she sees. Which of the following psychological defense mechanisms is she demonstrating?

Answer Choices:

A. Reaction formation B. Acting out C. Sexualization D. Intellectualization "

Correct Answer: A. Reaction formation

Question 128

Clinical Scenario:

A 34-year-old woman comes to the physician with fever and malaise. For the past 2 days, she has felt fatigued and weak and has had chills. Last night, she had a temperature of 40.8°C (104.2°F). She has also had difficulty swallowing since this morning. The patient was recently diagnosed with Graves disease and started on methimazole. She appears uncomfortable. Her temperature is 38.3°C (100.9°F), pulse is 95/min, and blood pressure is 134/74 mm Hg. The oropharynx is erythematous without exudate. The lungs are clear to auscultation. Laboratory studies show: Hematocrit 42% Hemoglobin 13.4 g/dL Leukocyte count 3,200/mm3 Segmented neutrophils 9% Basophils < 1% Eosinophils < 1% Lymphocytes 79% Monocytes 11% Platelet count 230,000/mm3 Which of the following is the most appropriate next step in management?"

Answer Choices:

A. Bone marrow biopsy B. Discontinue methimazole C. Test for EBV, HIV, and CMV D. Decrease methimazole dose

Correct Answer: B. Discontinue methimazole

Question 129

Clinical Scenario:

A 26-year-old woman, gravida 2, para 1, at 26 weeks' gestation, comes to the emergency department because of pain and swelling in her right calf. Physical examination shows an increased circumference of the right calf. The leg is warm and tender on palpation. Dorsiflexion of the right foot elicits calf pain. An ultrasound of the right leg shows a noncompressible popliteal vein. Which of the following is the most appropriate pharmacotherapy for this patient's condition?

Answer Choices:

A. Aspirin B. Clopidogrel C. Heparin D. Warfarin

Correct Answer: C. Heparin

Question 130

Clinical Scenario:

A 48-year-old man with a history of nephrolithiasis presents with acute-onset left flank pain. He says that the pain started suddenly 4 hours ago and has progressively worsened. He describes the pain as severe, sharp, and localized to the left flank. The patient denies any fever, chills, nausea, vomiting, or dysuria. His past medical history is significant for nephrolithiasis diagnosed 4 years ago status post shockwave lithotripsy. The patient says, "I'm allergic to many pain medications, but there is one that I get all the time when I have this pain. I think it starts with D". He is afebrile and his vital signs are stable. On physical examination, he is writhing in pain and moaning. Exquisite left costovertebral angle tenderness is noted. Laboratory findings, including a urinalysis, are unremarkable. IV fluid resuscitation is administered. Which of the following is the best next step in the management of this patient?

Answer Choices:

A. Admit to hospital floor for IV dilaudid patient-controlled analgesia B. Administer ibuprofen and acetaminophen for pain control C. Discharge patient with prescription of dilaudid with follow-up in 3 months D. Non-contrast CT of the abdomen and pelvis

Correct Answer: D. Non-contrast CT of the abdomen and pelvis

Question 131

Clinical Scenario:

A 3-month-old girl is brought to the emergency department in respiratory distress after her parents noticed that she was having difficulty breathing. They say that she developed a fever 2 days ago and subsequently developed increasing respiratory difficulty, lethargy, and productive cough. On presentation, her temperature is 103°F (39.5°C), blood pressure is 84/58 mmHg, pulse is 141/min, and respirations are 48/min. Physical exam reveals subcostal retractions and consolidation in the right lower lung field. She is also found to have coarse facial features and restricted joint movement. Serum laboratory tests reveal abnormally elevated levels of lysosomal enzymes circulating in the blood. The enzyme that is most likely defective in this patient has which of the following substrates?

Answer Choices:

A. Ceremide B. Dermatan sulfate C. Galactocerebroside D. Mannose

Correct Answer: D. Mannose

Question 132

Clinical Scenario:

A 1-minute-old newborn is being examined by the pediatric nurse. The nurse auscultates the heart and determines that the heart rate is 89/min. The respirations are spontaneous and regular. The chest and abdomen are both pink while the tips of the fingers and toes are blue. When the newborn's foot is slapped the face grimaces and he cries loud and strong. When the arms are extended by the nurse they flex back quickly. What is this patient's Apgar score?

Answer Choices:

A. 6 B. 8 C. 9 D. 10

Correct Answer: B. 8

Question 133

Clinical Scenario:

A 7-year-old boy presents to an urgent care clinic from his friend's birthday party after experiencing trouble breathing. His father explains that the patient had eaten peanut butter at the party, and soon after, he developed facial flushing and began scratching his face and neck. This has never happened before but his father says that they have avoided peanuts and peanut butter in the past because they were worried about their son having an allergic reaction. The patient has no significant medical history and takes no medications. His blood pressure is 94/62 mm Hg, heart rate is 125/min, and respiratory rate is 22/min. On physical examination, his lips are edematous and he has severe audible stridor. Of the following, which type of hypersensitivity reaction is this patient experiencing?

Answer Choices:

A. Type I hypersensitivity reaction B. Type II hypersensitivity reaction C. Type III hypersensitivity reaction D. Type IV hypersensitivity reaction

Correct Answer: A. Type I hypersensitivity reaction

Question 134

Clinical Scenario:

A 56-year-old woman comes to the emergency department because of a 3-day history of malaise, dysuria, blurred vision, and a painful, itchy rash. The rash began on her chest and face and spread to her limbs, palms, and soles. One week ago, she was diagnosed with trigeminal neuralgia and started on a new medicine. She appears ill. Her temperature is 38°C (100.4°F) and pulse is 110/min. Physical examination shows conjunctival injection and ulceration on the tongue and palate. There is no lymphadenopathy. Examination of the skin shows confluent annular, erythematous macules, bullae, and desquamation of the palms and soles. The epidermis separates when the skin is lightly stroked. Which of the following is the most likely diagnosis?

Answer Choices:

A. Bullous pemphigoid B. Stevens-Johnson syndrome C. Pemphigus vulgaris D. Drug-induced lupus erythematosus

Correct Answer: B. Stevens-Johnson syndrome

Question 135

Clinical Scenario:

A well-dressed couple presents to the emergency department with sudden onset of headache, a sensation of floating, and weakness of arms and legs after eating a plate of shellfish 2 hours ago. They mention that they had experienced tingling of the lips and mouth within 15 minutes of ingesting the shellfish. They also complain of mild nausea and abdominal discomfort. On physical examination, their vital signs are within normal limits. Their neurological examination reveals decreased strength in all extremities bilaterally and hyporeflexia. After detailed laboratory evaluation, the physician confirms the diagnosis of paralysis due to the presence of a specific toxin in the shellfish they had consumed. Which of the following mechanisms best explains the action of the toxin these patients had consumed?

Answer Choices:

A. Inactivation of syntaxin B. Blockade of voltage-gated fast sodium channels C. Increased opening of presynaptic calcium channels D. Inhibition of acetylcholinesterase

Correct Answer: B. Blockade of voltage-gated fast sodium channels

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Question 136

Clinical Scenario:

A 3-year-old boy is brought to the physician because of a 3-day history of fever and chills. The mother reports that he has also been limping for 2 days. He has no history of trauma to this region. His temperature is 38.9°C (102°F). Physical examination shows dull tenderness over his left lower extremity. The range of motion of the leg is also limited by pain. While walking, he avoids putting weight on his left leg. Laboratory studies show an erythrocyte sedimentation rate of 67 mm/h. An MRI is most likely to show abnormalities in which of the following regions?

Answer Choices:

A. Proximal metaphysis of the femur B. Proximal epiphysis of the femur C. Diaphysis of the tibia D. Acetabulum of the ilium

Correct Answer: A. Proximal metaphysis of the femur

Question 137

Clinical Scenario:

An autopsy of a patient's heart who recently died in a motor vehicle accident shows multiple nodules near the line of closure on the ventricular side of the mitral valve leaflet. Microscopic examination shows that these nodules are composed of immune complexes, mononuclear cells, and thrombi interwoven with fibrin strands. These nodules are most likely to be found in which of the following patients?

Answer Choices:

A. A 71-year-old male with acute-onset high fever and nail bed hemorrhages B. A 41-year-old female with a facial rash and nonerosive arthritis C. A 62-year-old male with Cardiobacterium hominis bacteremia D. A 6-year-old female with subcutaneous nodules and erythema marginatum

Correct Answer: B. A 41-year-old female with a facial rash and nonerosive arthritis

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Question 138

Clinical Scenario:

A 40-year-old male accountant is brought to the physician by his wife. She complains of her husband talking strangely for the past 6 months. She has taken him to multiple physicians during this time, but her husband did not comply with their treatment. She says he keeps things to himself, stays alone, and rarely spends time with her or the kids. When asked how he was doing, he responds in a clear manner with "I am fine, pine, dine doc." When further questioned about what brought him in today, he continues “nope, pope, dope doc.” Physical examination reveals no sensorimotor loss or visual field defects. Which of the following best describes the patient's condition?

Answer Choices:

A. It is associated with a better prognosis B. Patient has no insight C. Patient has disorganized thinking D. Confrontational psychoeducation would be beneficial

Correct Answer: C. Patient has disorganized thinking

Question 139

Clinical Scenario:

A 72-year-old man presents to his primary care physician complaining of pain and bulging in his groin. He is otherwise healthy and has never had surgery. He is referred to a general surgeon, and is scheduled for an elective hernia repair the following week. On closer inspection in the operating room, the surgeon notes a hernia sac that protrudes through the external inguinal ring, bypassing the inguinal canal. Which of the following accurately describes this patient's condition?

Answer Choices:

A. Indirect femoral hernia B. Direct incisional hernia C. Isolated rectus diastasis D. Direct inguinal hernia

Correct Answer: D. Direct inguinal hernia

Question 140

Clinical Scenario:

A 58-year-old woman presents to her physician complaining of a headache in the occipital region for 1 week. Past medical history is significant for essential hypertension, managed with lifestyle modifications and 2 antihypertensives for the previous 6 months. Her blood pressure is 150/90 mm Hg. Neurological examination is normal. A third antihypertensive drug is added that acts as a selective α2 adrenergic receptor agonist. On follow-up, she reports that she does not have any symptoms and her blood pressure is 124/82 mm Hg. Which of the following mechanisms best explains the therapeutic effect of this new drug in this patient?

Answer Choices:

A. Negative inotropic effect on the heart B. Vasodilation of peripheral veins C. Vasodilation of peripheral arteries D. Decreased peripheral sympathetic outflow

Correct Answer: D. Decreased peripheral sympathetic outflow

Question 141

Clinical Scenario:

A 50-year-old woman presents with altered taste and a gritty sensation in her eyes for the last month. She mentions that she needs to drink water frequently and often feels that her mouth and throat are dry. On physical examination, she has bilateral enlargement of the parotid glands and dry conjunctivae. Her physical examination and laboratory findings suggest a diagnosis of sicca syndrome. In addition to non-pharmacological measures, a drug is prescribed to improve symptoms related to dryness of mouth by increasing salivation. Which of the following is the mechanism of action of the drug that most likely is being prescribed to this patient?

Answer Choices:

A. Selective M1 muscarinic receptor antagonist B. Selective M2 muscarinic receptor agonist C. Selective M2 muscarinic receptor antagonist D. Selective M3 muscarinic receptor agonist

Correct Answer: D. Selective M3 muscarinic receptor agonist

Question 142

Clinical Scenario:

A six year-old female presents for evaluation of dry skin, fatigue, sensitivity to cold and constipation. The patient’s mother recalls that the patient had surgery to remove a “benign mass” at the base of her tongue 3 months ago because of trouble swallowing. What was the likely cause of the surgically removed mass?

Answer Choices:

A. Radiation exposure B. Iodine deficiency C. Failed caudal migration of the thyroid gland D. Failed fusion of the palatine shelves with the nasal septum

Correct Answer: C. Failed caudal migration of the thyroid gland

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Question 143

Clinical Scenario:

A 40-year-old homeless man is brought to the emergency department after police found him in the park lying on the ground with a minor cut at the back of his head. He is confused with slurred speech and fails a breathalyzer test. Pupils are normal in size and reactive to light. A bolus of intravenous dextrose, thiamine, and naloxone is given in the emergency department. The cut on the head is sutured. Blood and urine are drawn for toxicology screening. The blood-alcohol level comes out to be 200 mg/dL. Liver function test showed an AST of 320 U/L, ALT of 150 U/L, gamma-glutamyl transferase of 100 U/L, and total and direct bilirubin level are within normal limits. What is the most likely presentation with a person of this history?

Answer Choices:

A. Ataxic gait B. Pin point pupil C. Vertical nystagmus D. High blood pressure

Correct Answer: A. Ataxic gait

Question 144

Clinical Scenario:

A 38-year-old man presents with sudden onset abdominal pain and undergoes an emergent laparoscopic appendectomy. The procedure is performed quickly, without any complications, and the patient is transferred to the post-operative care unit. A little while later, the patient complains of seeing people in his room and hearing voices talking to him. The patient has no prior medical or psychiatric history and does not take any regular medications. What is the mechanism of action of the anesthetic most likely responsible for this patient’s symptoms?

Answer Choices:

A. Increased duration of GABA-gated chloride channel opening B. N-methyl-D-aspartate receptor antagonism C. Stimulation of μ-opioid receptors D. Blocking the fast voltage-gated Na+ channels

Correct Answer: B. N-methyl-D-aspartate receptor antagonism

Question 145

Clinical Scenario:

A clinical trial is being run with patients that have a genetic condition characterized by abnormal hemoglobin that can undergo polymerization when exposed to hypoxia, acidosis, or dehydration. This process of polymerization is responsible for the distortion of the red blood cell (RBC) that acquires a crescent shape and the hemolysis of RBCs. Researchers are studying the mechanisms of the complications commonly observed in these patients such as stroke, aplastic crisis, and auto-splenectomy. What kind of mutation leads to the development of the disease?

Answer Choices:

A. Missense mutation B. Splice site C. Frameshift mutation D. Silent mutation

Correct Answer: A. Missense mutation

Question 146

Clinical Scenario:

A 60-year-old woman comes to the physician because of lower back pain, generalized weakness, and weight loss that has occurred over the past 6 weeks. She also says that her urine has appeared foamy recently. Physical examination shows focal midline tenderness of the lumbar spine and conjunctival pallor. Her temperature is 100.5°F (38°C). A photomicrograph of a bone marrow biopsy specimen is shown. Further evaluation of this patient is most likely to show which of the following findings?

Answer Choices:

A. B-lymphocytes with radial cytoplasmic projections B. Neutrophils with hypersegmented nuclear lobes C. Grouped erythrocytes with stacked-coin appearance D. Myeloblasts with needle-shaped cytoplasmic inclusions

Correct Answer: C. Grouped erythrocytes with stacked-coin appearance

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Question 147

Clinical Scenario:

The drug cilostazol is known for its ability to relax vascular smooth muscle and therefore cause vasodilation through its inhibition of phosphodiesterase 3. Given this mechanism of action, what other effect would be expected?

Answer Choices:

A. Increased left ventricular end-diastolic volume B. Positive inotropy C. Angioedema D. Antiarrhythmic action

Correct Answer: B. Positive inotropy

Question 148

Clinical Scenario:

A 38-year-old man comes to the physician because of persistent sadness and difficulty concentrating for the past 6 weeks. During this period, he has also had difficulty sleeping. He adds that he has been “feeling down” most of the time since his girlfriend broke up with him 4 years ago. Since then, he has only had a few periods of time when he did not feel that way, but none of these lasted for more than a month. He reports having no problems with appetite, weight, or energy. He does not use illicit drugs or alcohol. Mental status examination shows a depressed mood and constricted affect. Which of the following is the most likely diagnosis?

Answer Choices:

A. Major depressive disorder B. Cyclothymic disorder C. Persistent depressive disorder D. Adjustment disorder with depressed mood

Correct Answer: C. Persistent depressive disorder

Question 149

Clinical Scenario:

A 40-year-old man comes to the physician because of a 1-month history of a painless lump on his neck. Two years ago, he underwent surgery for treatment-resistant hypertension, episodic headaches, and palpitations. Physical examination shows a firm, irregular swelling on the right side of the neck. Ultrasonography of the thyroid gland shows a 2-cm nodule with irregular margins and microcalcifications in the right thyroid lobe. Further evaluation of this patient is most likely to show increased serum concentration of which of the following substances?

Answer Choices:

A. Calcitonin B. Gastrin C. Metanephrines D. Thyroid-stimulating hormone

Correct Answer: A. Calcitonin

Question 150

Clinical Scenario:

A 22-year-old female presents to her PCP after having unprotected sex with her boyfriend 2 days ago. She has been monogamous with her boyfriend but is very concerned about pregnancy. The patient requests emergency contraception to decrease her likelihood of getting pregnant. A blood hCG test returns negative. The PCP prescribes the patient ethinyl estradiol 100 mcg and levonorgestrel 0.5 mg to be taken 12 hours apart. What is the most likely mechanism of action for this combined prescription?

Answer Choices:

A. Inhibition or delayed ovulation B. Thickening of cervical mucus with sperm trapping C. Tubal constriction inhibiting sperm transportation D. Interference of corpus luteum function

Correct Answer: A. Inhibition or delayed ovulation

Question 151

Clinical Scenario:

An infant boy of unknown age and medical history is dropped off in the emergency department. The infant appears lethargic and has a large protruding tongue. Although the infant exhibits signs of neglect, he is in no apparent distress. The heart rate is 70/min, the respiratory rate is 30/min, and the temperature is 35.7°C (96.2°F). Which of the following is the most likely cause of the patient's physical exam findings?

Answer Choices:

A. Congenital agenesis of an endocrine gland in the anterior neck B. Excess growth hormone secondary to pituitary gland tumor C. Type I hypersensitivity reaction D. Autosomal dominant mutation in the SERPING1 gene

Correct Answer: A. Congenital agenesis of an endocrine gland in the anterior neck

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Question 152

Clinical Scenario:

A 40-year-old man presents with problems with his vision. He says he has been experiencing blurred vision and floaters in his left eye for the past few days. He denies any ocular pain, fever, or headaches. Past medical history is significant for HIV infection a few years ago, for which he is noncompliant with his antiretroviral medications and his most recent CD4 count was 100 cells/mm3. His temperature is 36.5°C (97.7°F), the blood pressure is 110/89 mm Hg, the pulse rate is 70/min, and the respiratory rate is 14/min. Ocular exam reveals a decreased vision in the left eye, and a funduscopic examination is shown in the image. The patient is admitted and immediately started on intravenous ganciclovir. A few days after admission he is still complaining of blurry vision and floaters, so he is switched to a different medication. Inhibition of which of the following processes best describes the mechanism of action of the newly added medication?

Answer Choices:

A. Viral penetration into host cells B. Nucleic acid synthesis C. Progeny virus release D. Viral uncoating

Correct Answer: B. Nucleic acid synthesis

Question 153

Clinical Scenario:

A 5-week-old male infant is brought to the physician by his mother because of a 4-day history of recurrent nonbilious vomiting after feeding. He was born at 36 weeks' gestation via spontaneous vaginal delivery. Vital signs are within normal limits. Physical examination shows a 2-cm epigastric mass. Further diagnostic evaluation of this patient is most likely to show which of the following?

Answer Choices:

A. High serum 17-hydroxyprogesterone concentration B. Dilated colon segment on abdominal x-ray C. Elongated and thickened pylorus on abdominal ultrasound D. Corkscrew sign on upper gastrointestinal contrast series "

Correct Answer: C. Elongated and thickened pylorus on abdominal ultrasound

Question 154

Clinical Scenario:

A newborn male, delivered by emergency Cesarean section during the 28th week of gestation, has a birth weight of 1.2 kg (2.5 lb). He develops rapid breathing 4 hours after birth. Examination of the respiratory system reveals a respiratory rate of 80/min, expiratory grunting, intercostal and subcostal retractions with nasal flaring. His chest radiograph shows bilateral diffuse reticulogranular opacities and poor lung expansion. His echocardiography suggests a diagnosis of patent ductus arteriosus with left-to-right shunt and signs of fluid overload. The pediatrician administers intravenous indomethacin to facilitate closure of the duct. Which of the following effects best explains the mechanism of action of this drug in the management of this neonate?

Answer Choices:

A. Inhibition of lipoxygenase B. Increased synthesis of prostaglandin E2 C. Decreased blood flow in the vasa vasorum of the ductus arteriosus D. Increased synthesis of platelet-derived growth factor (PDGF)

Correct Answer: C. Decreased blood flow in the vasa vasorum of the ductus arteriosus

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Question 155

Clinical Scenario:

A 25-year-old G2P1001 at 32 weeks gestation presents to the hospital with painless vaginal bleeding. The patient states that she was taking care of laundry at home when she experienced a sudden sensation of her water breaking and saw that her groin was covered in blood. Her prenatal history is unremarkable according to the clinic records, but she has not seen an obstetrician for the past 14 weeks. Her previous delivery was by urgent cesarean section for placenta previa. Her temperature is 95°F (35°C), blood pressure is 125/75 mmHg, pulse is 79/min, respirations are 18/min, and oxygen saturation is 98% on room air. Cervical exam shows gross blood in the vaginal os. The fetal head is not palpable. Fetal heart rate monitoring demonstrates decelerations and bradycardia. Labs are pending. IV fluids are started. What is the best next step in management?

Answer Choices:

A. Betamethasone B. Cesarean section C. Lumbar epidural block D. Red blood cell transfusion

Correct Answer: B. Cesarean section

Question 156

Clinical Scenario:

A 52-year-old man is brought to the emergency department by ambulance after a motor vehicle accident. He was an unrestrained passenger who was ejected from the vehicle. On presentation, he is found to be actively bleeding from numerous wounds. His blood pressure is 76/42 mmHg and pulse is 152/min. Attempts at resuscitation fail, and he dies 25 minutes later. Autopsy shows blood in the peritoneal cavity, and histology of the kidney reveals swelling of the proximal convoluted tubule epithelial cells. Which of the following is most likely the mechanism underlying the renal cell findings?

Answer Choices:

A. Decreased activity of caspase 7 B. Decreased function of the Na+/K+-ATPase C. Increased activity of caspase 9 D. Increased function of the Na+/K+-ATPase

Correct Answer: B. Decreased function of the Na+/K+-ATPase

Question 157

Clinical Scenario:

A 37-year-old woman comes to the physician because of a 6-month history of weight loss, bloating, and diarrhea. She does not smoke or drink alcohol. Her vital signs are within normal limits. She is 173 cm (5 ft 8 in) tall and weighs 54 kg (120 lb); BMI is 18 kg/m2. Physical examination shows bilateral white spots on the temporal half of the conjunctiva, dry skin, and a hard neck mass in the anterior midline that does not move with swallowing. Urinalysis after a D-xylose meal shows an increase in renal D-xylose excretion. Which of the following is most likely to have prevented this patient's weight loss?

Answer Choices:

A. Gluten-free diet B. Pancreatic enzyme replacement C. Tetracycline therapy D. Lactose-free diet

Correct Answer: B. Pancreatic enzyme replacement

Question 158

Clinical Scenario:

A researcher is studying physiologic and hormonal changes that occur during pregnancy. Specifically, they examine the behavior of progesterone over the course of the menstrual cycle and find that it normally decreases over time; however, during pregnancy this decrease does not occur in the usual time frame. The researcher identifies a circulating factor that appears to be responsible for this difference in progesterone behavior. In order to further examine this factor, the researcher denatures the circulating factor and examines the sizes of its components on a western blot as compared to several other hormones. One of the bands the researcher identifies in this circulating factor is identical to that of another known hormone with which of the following sites of action?

Answer Choices:

A. Adipocytes B. Adrenal gland C. Bones D. Thyroid gland

Correct Answer: D. Thyroid gland

Question 159

Clinical Scenario:

A 53-year-old man with a past medical history significant for hyperlipidemia, hypertension, and hyperhomocysteinemia presents to the emergency department complaining of 10/10 crushing, left-sided chest pain radiating down his left arm and up his neck into the left side of his jaw. His ECG shows ST-segment elevation in leads V2-V4. He is taken to the cardiac catheterization laboratory for successful balloon angioplasty and stenting of a complete blockage in his left anterior descending coronary artery. Echocardiogram the following day shows decreased left ventricular function and regional wall motion abnormalities. A follow-up echocardiogram 14 days later shows a normal ejection fraction and no regional wall motion abnormalities. This post-infarct course illustrates which of the following concepts?

Answer Choices:

A. Ventricular remodeling B. Myocardial hibernation C. Myocardial stunning D. Coronary collateral circulation

Correct Answer: C. Myocardial stunning

Question 160

Clinical Scenario:

A 25-year-old woman comes to the physician because of a 4-month history of anxiety and weight loss. She also reports an inability to tolerate heat and intermittent heart racing for 2 months. She appears anxious. Her pulse is 108/min and blood pressure is 145/87 mm Hg. Examination shows a fine tremor of her outstretched hands. After confirmation of the diagnosis, the patient is scheduled for radioactive iodine ablation. At a follow-up visit 2 months after the procedure, she reports improved symptoms but new-onset double vision. Examination shows conjunctival injections, proptosis, and a lid lag. Slit-lamp examination shows mild corneal ulcerations. The patient is given an additional medication that improves her diplopia and proptosis. Which of the following mechanisms is most likely responsible for the improvement in this patient's ocular symptoms?

Answer Choices:

A. Inhibition of iodide oxidation B. Elimination of excess fluid C. Replacement of thyroid hormones D. Decreased production of proinflammatory cytokines

Correct Answer: D. Decreased production of proinflammatory cytokines

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Question 161

Clinical Scenario:

A 19-year-old woman presents to her university health clinic for a regularly scheduled visit. She has a past medical history of depression, acne, attention-deficit/hyperactivity disorder, and dysmenorrhea. She is currently on paroxetine, dextroamphetamine, and naproxen during her menses. She is using nicotine replacement products to quit smoking. She is concerned about her acne, recent weight gain, and having a depressed mood this past month. She also states that her menses are irregular and painful. She is not sexually active and tries to exercise once a month. Her temperature is 97.6°F (36.4°C), blood pressure is 133/81 mmHg, pulse is 80/min, respirations are 14/min, and oxygen saturation is 98% on room air. Physical exam is notable for a morbidly obese woman with acne on her face. Her pelvic exam is unremarkable. The patient is given a prescription for isotretinoin. Which of the following is the most appropriate next step in management?

Answer Choices:

A. Check hCG B. Check prolactin C. Check TSH D. Recheck blood pressure in 1 week

Correct Answer: A. Check hCG

Question 162

Clinical Scenario:

A 54-year-old woman appears in your office for a new patient visit. She reports a past medical history of hypertension, which she was told was related to "adrenal gland disease." You recall that Conn syndrome and pheochromocytomas are both conditions affecting the adrenal gland that result in hypertension by different mechanisms. Which areas of the adrenal gland are involved in Conn syndrome and pheochromocytomas, respectively?

Answer Choices:

A. Zona glomerulosa; zona fasciculata B. Zona glomerulosa; medulla C. Medulla; zona reticularis D. Zona fasciculata; zona reticularis

Correct Answer: B. Zona glomerulosa; medulla

Question 163

Clinical Scenario:

A 2-year-old boy has a history of recurrent bacterial infections, especially of his skin. When he has an infection, pus does not form. His mother reports that, when he was born, his umbilical cord took 5 weeks to detach. He is ultimately diagnosed with a defect in a molecule in the pathway that results in neutrophil extravasation. Which of the following correctly pairs the defective molecule with the step of extravasation that molecule affects?

Answer Choices:

A. ICAM-1; margination B. LFA-1 (integrin); margination C. LFA-1 (integrin); tight adhesion D. E-selectin; tight adhesion

Correct Answer: C. LFA-1 (integrin); tight adhesion

Question 164

Clinical Scenario:

A 34-year-old woman with Crohn disease comes to the physician because of a 4-week history of nausea, bloating, and epigastric pain that occurs after meals and radiates to the right shoulder. Four months ago, she underwent ileocecal resection for an acute intestinal obstruction. An ultrasound of the abdomen shows multiple echogenic foci with acoustic shadows in the gallbladder. Which of the following mechanisms most likely contributed to this patient's current presentation?

Answer Choices:

A. Increased hepatic cholesterol secretion B. Decreased fat absorption C. Decreased motility of the gallbladder D. Decreased biliary concentration of bile acids

Correct Answer: D. Decreased biliary concentration of bile acids

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Question 165

Clinical Scenario:

A 72-year-old male presents to his primary care physician complaining of increased urinary frequency and a weakened urinary stream. He has a history of gout, obesity, diabetes mellitus, and hyperlipidemia. He currently takes allopurinol, metformin, glyburide, and rosuvastatin. His temperature is 98.6°F (37°C), blood pressure is 130/85 mmHg, pulse is 90/min, and respirations are 18/min. Physical examination reveals an enlarged, non-tender prostate without nodules or masses. An ultrasound reveals a uniformly enlarged prostate that is 40mL in size. His physician starts him on a new medication. After taking the first dose, the patient experiences lightheadedness upon standing and has a syncopal event. Which of the following mechanisms of action is most consistent with the medication in question?

Answer Choices:

A. Alpha-1-adrenergic receptor antagonist B. Alpha-2-adrenergic receptor agonist C. Non-selective alpha receptor antagonist D. Selective muscarinic agonist

Correct Answer: A. Alpha-1-adrenergic receptor antagonist

Question 166

Clinical Scenario:

A 22-year-old man has had dyspnea and hemoptysis for the past week. He has no known sick contacts. There is no personal or family history of serious illness. He takes no medications. His temperature is 37°C (98.6°F), pulse is 82/min, respirations are 22/min, and blood pressure is 152/90 mm Hg. Examination shows inspiratory crackles at both lung bases. The remainder of the examination shows no abnormalities. His hemoglobin is 14.2 g/dL, leukocyte count is 10,300/mm3, and platelet count is 205,000/mm3. Urinalysis shows a proteinuria of 2+, 70 RBC/hpf, and 1–2 WBC/hpf. Chest x-ray shows pulmonary infiltrates. Further evaluation is most likely to show which of the following findings?

Answer Choices:

A. Increased anti-GBM antibody titers B. Increased c-ANCA titers C. Increased p-ANCA titers D. Increased anti-dsDNA antibody titers

Correct Answer: A. Increased anti-GBM antibody titers

Question 167

Clinical Scenario:

A young woman from the Ohio River Valley in the United States currently on corticosteroid therapy for ulcerative colitis presented to a clinic complaining of fever, sweat, headache, nonproductive cough, malaise, and general weakness. A chest radiograph revealed patchy pneumonia in the lower lung fields, together with enlarged mediastinal and hilar lymph nodes. Skin changes suggestive of erythema nodosum (i.e. an acute erythematous eruption) were noted. Because the patient was from a region endemic for fungal infections associated with her symptoms and the patient was in close contact with a person presenting similar symptoms, the attending physician suspected that systemic fungal infection might be responsible for this woman's illness. Which of the following laboratory tests can the physician use to ensure early detection of the disease, and also effectively monitor the treatment response?

Answer Choices:

A. Culture method B. Antibody testing C. Fungal staining D. Antigen detection

Correct Answer: D. Antigen detection

Question 168

Clinical Scenario:

A 55-year-old man comes to the physician with a 3-month history of headache, periodic loss of vision, and easy bruising. Physical examination shows splenomegaly. His hemoglobin concentration is 13.8 g/dL, leukocyte count is 8000/mm3, and platelet count is 995,000/mm3. Bone marrow biopsy shows markedly increased megakaryocytes with hyperlobulated nuclei. Genetic analysis shows upregulation of the JAK-STAT genes. The pathway encoded by these genes is also physiologically responsible for signal transmission of which of the following hormones?

Answer Choices:

A. Cortisol B. Oxytocin C. Prolactin D. Adrenocorticotropic hormone

Correct Answer: C. Prolactin

Question 169

Clinical Scenario:

An investigator is studying the effect that mutations in different parts of the respiratory tract have on susceptibility to infection. A mutation in the gene encoding for the CD21 protein is induced in a sample of cells obtained from the nasopharyngeal epithelium. This mutation is most likely to prevent infection with which of the following viruses?

Answer Choices:

A. Rhinovirus B. Epstein-Barr virus C. Cytomegalovirus D. Parvovirus

Correct Answer: B. Epstein-Barr virus

Question 170

Clinical Scenario:

A 43-year-old Hispanic woman was admitted to the emergency room with intermittent sharp and dull pain in the right lower quadrant for the past 2 days. The patient denies nausea, vomiting, diarrhea, or fever. She states that she was 'completely normal' prior to this sudden episode of pain. The patient states that she is sure she is not currently pregnant and notes that she has no children. Physical exam revealed guarding on palpation of the lower quadrants. An abdominal ultrasound revealed free abdominal fluid, as well as fluid in the gallbladder fossa. After further evaluation, the patient is considered a candidate for laparoscopic cholecystectomy. The procedure and the risks of surgery are explained to her and she provides informed consent to undergo the cholecystectomy. During the procedure, the surgeon discovers a gastric mass suspicious for carcinoma. The surgeon considers taking a biopsy of the mass to determine whether or not she should resect the mass if it proves to be malignant. Which of the following is the most appropriate course of action to take with regards to taking a biopsy of the gastric mass?

Answer Choices:

A. The surgeon should resect the gastric mass B. The surgeon should obtain consent to biopsy the mass from the patient when she wakes up from cholecystectomy C. The surgeon should contact an ethics committee to obtain consent to biopsy the mass D. The surgeon should contact an attorney to obtain consent to biopsy the mass

Correct Answer: B. The surgeon should obtain consent to biopsy the mass from the patient when she wakes up from cholecystectomy

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Question 171

Clinical Scenario:

A 12-year-old male presents to the pediatrician after two days of tea-colored urine which appeared to coincide with the first day of junior high football. He explains that he refused to go back to practice because he was humiliated by the other players due to his quick and excessive fatigue after a set of drills accompanined by pain in his muscles. A blood test revealed elevated creatine kinase and myoglobulin levels. A muscle biopsy was performed revealing large glycogen deposits and an enzyme histochemistry showed a lack of myophosphorylase activity. Which of the following reactions is not occuring in this individuals?

Answer Choices:

A. Converting glucose-6-phosphate to glucose B. Breaking down glycogen to glucose-1-phosphate C. Cleaving alpha-1,6 glycosidic bonds from glycogen D. Converting galactose to galactose-1-phosphate

Correct Answer: B. Breaking down glycogen to glucose-1-phosphate

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Question 172

Clinical Scenario:

Maturity Onset Diabetes of the Young (MODY) type 2 is a consequence of a defective pancreatic enzyme, which normally acts as a glucose sensor, resulting in a mild hyperglycemia. The hyperglycemia is especially exacerbated during pregnancy. Which of the following pathways is controlled by this enzyme?

Answer Choices:

A. Glucose --> glucose-6-phosphate B. Glucose-6-phosphate --> fructose-6-phosphate C. Fructose-6-phosphate --> fructose-1,6-biphosphate D. Phosphoenolpyruvate --> pyruvate

Correct Answer: A. Glucose --> glucose-6-phosphate

Question 173

Clinical Scenario:

A 72-year-old African American man presents with progressive fatigue, difficulty breathing on exertion, and lower extremity swelling for 3 months. The patient was seen at the emergency department 2 times before. The first time was because of back pain, and the second was because of fever and cough. He took medications at the emergency room, but he refused to do further tests recommended to him. He does not smoke or drink alcohol. His family history is irrelevant. His vital signs include a blood pressure of 110/80 mm Hg, temperature of 37.2°C (98.9°F), and regular radial pulse of 90/min. On physical examination, the patient looks pale, and his tongue is enlarged. Jugular veins become distended on inspiration. Pitting ankle edema is present on both sides. Bilateral basal crackles are audible on the chest auscultation. Hepatomegaly is present on abdominal palpation. Chest X-ray shows osteolytic lesions of the ribs. ECG shows low voltage waves and echocardiogram shows a speckled appearance of the myocardium with diastolic dysfunction and normal appearance of the pericardium. Which of the following best describes the mechanism of this patient’s illness?

Answer Choices:

A. Deposition of an extracellular fibrillar protein that stains positive for Congo red in the myocardium B. Concentric hypertrophy of the myocytes with thickening of the interventricular septum C. Calcification of the aortic valve orifice with obstruction of the left ventricular outflow tract D. Diastolic cardiac dysfunction with reciprocal variation in ventricular filling with respiration

Correct Answer: A. Deposition of an extracellular fibrillar protein that stains positive for Congo red in the myocardium

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Question 174

Clinical Scenario:

A 62-year-old man comes to the physician because of a 5-day history of fatigue, fever, and chills. For the past 9 months, he has had hand pain and stiffness that has progressively worsened. He started a new medication for these symptoms 3 months ago. Medications used prior to that included ibuprofen, prednisone, and hydroxychloroquine. He does not smoke or drink alcohol. Examination shows a subcutaneous nodule at his left elbow, old joint destruction with boutonniere deformity, and no active joint warmth or tenderness. The remainder of the physical examination shows no abnormalities. His hemoglobin concentration is 10.5 g/dL, leukocyte count is 3500/mm3, and platelet count is 100,000/mm3. Which of the following is most likely to have prevented this patient's laboratory abnormalities?

Answer Choices:

A. Amifostine B. Pyridoxine C. Leucovorin D. Mesna "

Correct Answer: C. Leucovorin

Question 175

Clinical Scenario:

A group of researchers conducted various studies on hepatitis C incidence and prevalence. They noticed that there is a high prevalence of hepatitis C in third-world countries, where it has a significant impact on the quality of life of the infected individual. The research group made several attempts to produce a vaccine that prevents hepatitis C infection but all attempts failed. Which of the following would most likely be the reason for the failure to produce a vaccine?

Answer Choices:

A. Tolerance B. Antigenic variation C. Non-DNA genome D. Polysaccharide envelope

Correct Answer: B. Antigenic variation

Question 176

Clinical Scenario:

A 61-year-old female is referred to an oncologist for evaluation of a breast lump that she noticed two weeks ago while doing a breast self-examination. Her past medical history is notable for essential hypertension and major depressive disorder for which she takes lisinopril and escitalopram, respectively. Her temperature is 98.6°F (37°C), blood pressure is 120/65 mmHg, pulse is 82/min, and respirations are 18/min. Biopsy of the lesion confirms a diagnosis of invasive ductal carcinoma with metastatic disease in the ipsilateral axillary lymph nodes. The physician starts the patient on a multi-drug chemotherapeutic regimen. The patient successfully undergoes mastectomy and axillary dissection and completes the chemotherapeutic regimen. However, several months after completion of the regimen, the patient presents to the emergency department with dyspnea, chest pain, and palpitations. A chest radiograph demonstrates an enlarged cardiac silhouette. This patient’s current symptoms could have been prevented by administration of which of the following medications?

Answer Choices:

A. Dexrazoxane B. Aspirin C. Rosuvastatin D. Cyclophosphamide

Correct Answer: A. Dexrazoxane

Question 177

Clinical Scenario:

A 16-year-old boy is brought to the physician by his mother because she is worried about his behavior. Yesterday, he was expelled from school for repeatedly skipping classes. Over the past 2 months, he was suspended 3 times for bullying and aggressive behavior towards his peers and teachers. Once, his neighbor found him smoking cigarettes in his backyard. In the past, he consistently maintained an A grade average and had been a regular attendee of youth group events at their local church. The mother first noticed this change in behavior 3 months ago, around the time at which his father moved out after discovering his wife was having an affair. Which of the following defense mechanisms best describes the change in this patient's behavior?

Answer Choices:

A. Acting out B. Projection C. Passive aggression D. Regression

Correct Answer: A. Acting out

Question 178

Clinical Scenario:

Two days after admission to the hospital, a 74-year-old man develops confusion and headache. He has also been vomiting over the past hour. His temperature is 36.7°C (98°F), pulse is 98/min, respirations are 22/min, and blood pressure is 140/80 mm Hg. He is lethargic and oriented only to person. Examination shows flushed skin. Fundoscopic examination shows bright red retinal veins. Serum studies show: Na^+ 138 mEq/L K^+ 3.5 mEq/L Cl^- 100 mEq/L HCO_3^- 17 mEq/L Creatinine 1.2 mg/dL Urea nitrogen 19 mg/dL Lactate 8.0 mEq/L (N = 0.5 - 2.2 mEq/L) Glucose 75 mg/dL Arterial blood gas analysis on room air shows a pH of 7.13. This patient's current presentation is most likely due to treatment for which of the following conditions?"

Answer Choices:

A. Hypertensive crisis B. Tension headache C. Major depressive disorder D. Acute dystonia

Correct Answer: A. Hypertensive crisis

Question 179

Clinical Scenario:

A 78-year-old man suffers a fall in a nursing home and is brought to the emergency room. A right hip fracture is diagnosed, and he is treated with a closed reduction with internal fixation under spinal anesthesia. On the second postoperative day, the patient complains of pain in the lower abdomen and states that he has not urinated since the surgery. An ultrasound shows increased bladder size and volume. Which of the following is the mechanism of action of the drug which is most commonly used to treat this patient’s condition?

Answer Choices:

A. Parasympathetic agonist B. Sympathetic agonist C. Alpha-blocker D. Beta-blocker

Correct Answer: A. Parasympathetic agonist

Question 180

Clinical Scenario:

A 36-year-old woman is brought to the emergency department after the sudden onset of severe, generalized abdominal pain. The pain is constant and she describes it as 9 out of 10 in intensity. She has hypertension, hyperlipidemia, and chronic lower back pain. Menses occur at regular 28-day intervals with moderate flow and last 4 days. Her last menstrual period was 2 weeks ago. She is sexually active with one male partner and uses condoms inconsistently. She has smoked one pack of cigarettes daily for 15 years and drinks 2–3 beers on the weekends. Current medications include ranitidine, hydrochlorothiazide, atorvastatin, and ibuprofen. The patient appears ill and does not want to move. Her temperature is 38.4°C (101.1°F), pulse is 125/min, respirations are 30/min, and blood pressure is 85/40 mm Hg. Examination shows a distended, tympanic abdomen with diffuse tenderness, guarding, and rebound; bowel sounds are absent. Her leukocyte count is 14,000/mm3 and hematocrit is 32%. Which of the following is the most likely cause of this patient's pain?

Answer Choices:

A. Ruptured ectopic pregnancy B. Bowel obstruction C. Perforation D. Colorectal cancer

Correct Answer: C. Perforation

Question 181

Clinical Scenario:

A previously healthy 21-year-old man is brought to the emergency department for the evaluation of an episode of unconsciousness that suddenly happened while playing football 30 minutes ago. He was not shaking and regained consciousness after about 30 seconds. Over the past three months, the patient has had several episodes of shortness of breath while exercising as well as sensations of a racing heart. He does not smoke or drink alcohol. He takes no medications. His vital signs are within normal limits. On mental status examination, he is oriented to person, place, and time. Cardiac examination shows a systolic ejection murmur that increases with valsalva maneuver and standing and an S4 gallop. The remainder of the examination shows no abnormalities. An ECG shows a deep S wave in lead V1 and tall R waves in leads V5 and V6. Echocardiography is most likely to show which of the following findings?

Answer Choices:

A. Abnormal movement of the mitral valve B. Ventricular septum defect C. Mitral valve leaflet thickening ≥ 5 mm D. Reduced left ventricular ejection fraction

Correct Answer: A. Abnormal movement of the mitral valve

Question 182

Clinical Scenario:

A neurophysiology expert is teaching his students the physiology of the neuromuscular junction. While describing the sequence of events that takes place at the neuromuscular junction, he mentions that as the action potential travels down the motor neuron, it causes depolarization of the presynaptic membrane. This results in the opening of voltage-gated calcium channels, which leads to an influx of calcium into the synapse of the motor neuron. Consequently, the cytosolic concentration of Ca2+ ions increases. Which of the following occurs at the neuromuscular junction as a result of this increase in cytosolic Ca2+?

Answer Choices:

A. Release of Ca2+ ions into the synaptic cleft B. Increased Na+ and K+ conductance of the motor end plate C. Exocytosis of acetylcholine from the synaptic vesicles D. Generation of an end plate potential

Correct Answer: C. Exocytosis of acetylcholine from the synaptic vesicles

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Question 183

Clinical Scenario:

Expression of an mRNA encoding for a soluble form of the Fas protein prevents a cell from undergoing programmed cell death. However, after inclusion of a certain exon, this same Fas pre-mRNA eventually leads to the translation of a protein that is membrane bound, subsequently promoting the cell to undergo apoptosis. Which of the following best explains this finding?

Answer Choices:

A. Base excision repair B. Histone deacetylation C. Post-translational modifications D. Alternative splicing

Correct Answer: D. Alternative splicing

Question 184

Clinical Scenario:

A 26-year-old G1P0 woman presents for her first prenatal visit. Past medical history reveals the patient is blood type O negative, and the father is type A positive. The patient refuses Rho(D) immune globulin (RhoGAM), because it is derived from human plasma, and she says she doesn't want to take the risk of contracting HIV. Which of the following is correct regarding the potential condition her baby may develop?

Answer Choices:

A. Rho(D) immune globulin is needed both before and immediately after delivery to protect this baby from developing the condition B. She should receive Rho(D) immune globulin to prevent the development of Rh(D) alloimmunization C. The Rho(D) immune globulin will also protect the baby against other Rh antigens aside from Rh(D) D. The injection can be avoided because the risk of complications of this condition is minimal

Correct Answer: B. She should receive Rho(D) immune globulin to prevent the development of Rh(D) alloimmunization

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Question 185

Clinical Scenario:

Three weeks after starting a new medication for hyperlipidemia, a 54-year-old man comes to the physician because of pain and swelling in his left great toe. Examination shows swelling and erythema over the metatarsophalangeal joint of the toe. Analysis of fluid from the affected joint shows needle-shaped, negatively-birefringent crystals. Which of the following best describes the mechanism of action of the drug he is taking?

Answer Choices:

A. Inhibition of hepatic HMG-CoA reductase B. Inhibition of intestinal bile acid absorption C. Inhibition of hepatic VLDL synthesis D. Inhibition of intestinal cholesterol absorption

Correct Answer: C. Inhibition of hepatic VLDL synthesis

Question 186

Clinical Scenario:

A 45-year-old woman with history of systemic sclerosis presents with new onset dyspnea, which is worsened with moderate exertion. She also complains of chest pain. An ECG was obtained, and showed right-axis deviation. Chest x-ray showed right ventricle hypertrophy. Given the patient's history and presentation, right heart catheterization was performed, which confirmed the suspected diagnosis of pulmonary artery hypertension. It is decided to start the patient on bosentan. Which of the following describes the method of action of bosentan?

Answer Choices:

A. Endothelin receptor antagonist B. Endothelin receptor agonist C. Phosphodiesterase type 5 inhibitor D. Calcium channel blocker

Correct Answer: A. Endothelin receptor antagonist

Question 187

Clinical Scenario:

A previously healthy 42-year-old woman comes to the physician because of a 7-month history of diffuse weakness. There is no cervical or axillary lymphadenopathy. Cardiopulmonary and abdominal examination shows no abnormalities. A lateral x-ray of the chest shows an anterior mediastinal mass. Further evaluation of this patient is most likely to show which of the following?

Answer Choices:

A. Acetylcholine receptor antibodies B. Elevated serum alpha-fetoprotein level C. History of fever, night sweats, and weight loss D. Increased urinary catecholamines

Correct Answer: A. Acetylcholine receptor antibodies

Question 188

Clinical Scenario:

A newborn is found to have cystic fibrosis during routine newborn screening. The parents, both biochemists, are curious about the biochemical basis of their newborn's condition. The pediatrician explains that the mutation causing cystic fibrosis affects the CFTR gene which codes for the CFTR channel. Which of the following correctly describes the pathogenesis of the most common CFTR mutation?

Answer Choices:

A. Insufficient CFTR channel production B. Defective post-translational glycosylation of the CFTR channel C. Defective post-translational hydroxylation of the CFTR channel D. Defective post-translational phosphorylation of the CFTR channel

Correct Answer: B. Defective post-translational glycosylation of the CFTR channel

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Question 189

Clinical Scenario:

A 24-year-old male graduate student presents to the physician for a 2-month history of persistent thoughts and anxiety that he is going to be harmed by someone on the street. The anxiety worsened after he witnessed a pedestrian being hit by a car 2 weeks ago. He states, “That was a warning sign.” On his way to school, he now often leaves an hour earlier to take a detour and hide from people that he thinks might hurt him. He is burdened by his coursework and fears that his professors are trying to fail him. He says his friends are concerned about him, but claims they do not understand because they were not present at the accident. The patient has no known history of any psychiatric illnesses. On the mental status exam, he is alert and oriented, and he shows a full range of affect. Thought processes and speech are organized. His memory and attention are within normal limits. He denies auditory, visual, or tactile hallucinations. The results of urine toxicology screening are negative. Which of the following is the most likely diagnosis in this patient?

Answer Choices:

A. Avoidant personality disorder B. Delusional disorder C. Generalized anxiety disorder D. Schizophrenia

Correct Answer: B. Delusional disorder

Question 190

Clinical Scenario:

A case-control study looking to study the relationship between infection with the bacterium Chlamydia trachomatis and having multiple sexual partners was conducted in the United States. A total of 100 women with newly diagnosed chlamydial infection visiting an outpatient clinic for sexually transmitted diseases (STDs) were compared with 100 women from the same clinic who were found to be free of chlamydia and other STDs. The women diagnosed with this infection were informed that the potential serious consequences of the disease could be prevented only by locating and treating their sexual partners. Both groups of women were queried about the number of sexual partners they had had during the preceding 3 months. The group of women with chlamydia reported an average of 4 times as many sexual partners compared with the group of women without chlamydia; the researchers, therefore, concluded that women with chlamydia visiting the clinic had significantly more sexual partners compared with women who visited the same clinic but were not diagnosed with chlamydia. What type of systematic error could have influenced the results of this study?

Answer Choices:

A. Ascertainment bias B. Response bias C. Detection bias D. Reporting bias

Correct Answer: D. Reporting bias

Question 191

Clinical Scenario:

A 4-year-old girl is brought to the physician because of increasing swelling around her eyes and over both her feet for the past 4 days. During this period, she has had frothy light yellow urine. Her vital signs are within normal limits. Physical examination shows periorbital edema and 2+ pitting edema of the lower legs and ankles. A urinalysis of this patient is most likely to show which of the following findings?

Answer Choices:

A. Muddy brown casts B. Epithelial casts C. Fatty casts D. WBC casts

Correct Answer: C. Fatty casts

Question 192

Clinical Scenario:

A 23-year-old woman comes to the physician because of a 2-month history of diarrhea, flatulence, and fatigue. She reports having 3–5 episodes of loose stools daily that have an oily appearance. The symptoms are worse after eating. She also complains of an itchy rash on her elbows and knees. A photograph of the rash is shown. Further evaluation of this patient is most likely to show which of the following findings?

Answer Choices:

A. Macrocytic, hypochromic red blood cells B. PAS-positive intestinal macrophages C. HLA-DQ2 serotype D. Elevated urine tryptophan levels

Correct Answer: C. HLA-DQ2 serotype

Question 193

Clinical Scenario:

A 25-year-old woman presents to the ED with nausea, vomiting, diarrhea, abdominal pain, and hematemesis after ingesting large quantities of a drug. Which of the following pairs a drug overdose with the correct antidote for this scenario?

Answer Choices:

A. Iron; deferoxamine B. Atropine; fomepizole C. Organophosphate; physostigmine D. Acetaminophen; naloxone

Correct Answer: A. Iron; deferoxamine

Question 194

Clinical Scenario:

A 42-year-old man is brought in to the emergency department by his daughter. She reports that her father drank heavily for the last 16 years, but he stopped 4 days ago after he decided to quit drinking on his birthday. She also reports that he has been talking about seeing cats running in his room since this morning, although there were no cats. There is no history of any known medical problems or any other substance use. On physical examination, his temperature is 38.4ºC (101.2ºF), heart rate is 116/min, blood pressure is 160/94 mm Hg, and respiratory rate is 22/min. He is severely agitated and is not oriented to his name, time, or place. On physical examination, profuse perspiration and tremors are present. Which of the following best describes the pathophysiologic mechanism underlying his condition?

Answer Choices:

A. Functional increase in GABA B. Increased activity of NMDA receptors C. Increased inhibition of norepinephrine D. Increased inhibition of glutamate

Correct Answer: B. Increased activity of NMDA receptors

Question 195

Clinical Scenario:

A 31-year-old female with a bacterial infection is prescribed a drug that binds the dipeptide D-Ala-D-Ala. Which of the following drugs was this patient prescribed?

Answer Choices:

A. Penicillin B. Chloramphenicol C. Vancomycin D. Polymyxin B

Correct Answer: C. Vancomycin

Question 196

Clinical Scenario:

A 24-year-old woman comes to her physician because of fatigue. She has been coming to the office multiple times a month for various minor problems over the past six months. During the appointments, she insists on a first name basis and flirts with her physician. She always dresses very fashionably. When his assistant enters the room, she tends to start fidgeting and interrupt their conversation. When the physician tells her politely that her behavior is inappropriate, she begins to cry, complaining that no one understands her and that if people only listened to her, she would not be so exhausted. She then quickly gathers herself and states that she will just have to keep looking for a physician who can help her, although she has doubts she will ever find the right physician. She does not have a history of self harm or suicidal ideation. Which of the following is the most likely diagnosis?

Answer Choices:

A. Dependent personality disorder B. Histrionic personality disorder C. Borderline personality disorder D. Schizotypal personality disorder

Correct Answer: B. Histrionic personality disorder

Question 197

Clinical Scenario:

A scientist is studying the mechanism by which the gastrointestinal system coordinates the process of food digestion. Specifically, she is interested in how distension of the lower esophagus by a bolus of food changes responses in the downstream segments of the digestive system. She observes that there is a resulting relaxation and opening of the lower esophageal (cardiac) sphincter after the introduction of a food bolus. She also observes a simultaneous relaxation of the orad stomach during this time. Which of the following substances is most likely involved in the process being observed here?

Answer Choices:

A. Ghrelin B. Neuropeptide-Y C. Secretin D. Vasoactive intestinal polypeptide

Correct Answer: D. Vasoactive intestinal polypeptide

Question 198

Clinical Scenario:

A 14-year-old boy comes to the physician for a follow-up after a blood test showed a serum triglyceride level of 821 mg/dL. Several of his family members have familial hypertriglyceridemia. The patient is prescribed a drug that increases his risk of gallstone disease. The expected beneficial effect of this drug is most likely due to which of the following actions?

Answer Choices:

A. Increased lipoprotein lipase activity B. Decreased lipolysis in adipose tissue C. Increased PPAR-gamma activity D. Increased bile acid sequestration

Correct Answer: A. Increased lipoprotein lipase activity

Question 199

Clinical Scenario:

A previously-healthy 24-year-old male is admitted to the intensive care unit following a motorcycle crash. He sustained head trauma requiring an emergency craniotomy, has burns over 30% of his body, and a fractured humerus. His pain is managed with a continuous fentanyl infusion. Two days after admission to the ICU he develops severe hematemesis. What is the mechanism underlying the development of his hematemesis?

Answer Choices:

A. Gastric mucosal disruption B. Increased gastric acid production C. Answers 1 and 2 D. Fentanyl overuse

Correct Answer: C. Answers 1 and 2

Question 200

Clinical Scenario:

A 37-year-old woman is brought to the physician for worsening depressive mood and irritability. Her mood changes began several months ago. Her husband has also noticed shaky movements of her limbs and trunk for the past year. The patient has no suicidal ideation. She has no history of serious illness and takes no medications. Her father died by suicide at the age of 45 years. Her temperature is 37°C (98.6°F), pulse is 76/min, and blood pressure is 128/72 mm Hg. She speaks slowly and quietly and only looks at the floor. She registers 3/3 words but can recall only one word 5 minutes later. Examination shows irregular movements of the arms and legs at rest. Extraocular eye movements are normal. Muscle strength is 5/5 throughout, and deep tendon reflexes are 2+ bilaterally. Further evaluation is most likely to show which of the following?

Answer Choices:

A. Mitral vegetations on echocardiogram B. Positive Babinski sign on physical examination C. Oligoclonal bands on lumbar puncture D. Caudate nucleus atrophy on MRI

Correct Answer: D. Caudate nucleus atrophy on MRI

Question 201

Clinical Scenario:

A 61-year-old woman comes to the physician because of a 6-day history of cough, shortness of breath, and fever. She also reports that she has had 4 episodes of watery diarrhea per day for the last 3 days. She has chronic bronchitis. She has smoked one pack of cigarettes daily for the past 30 years. Her temperature is 39°C (102.2°F) and pulse is 65/min. Examination shows diffuse crackles over the left lower lung field. Laboratory studies show: Hemoglobin 13.8 g/dL Leukocyte count 16,000/mm3 Platelet count 150,000/mm3 Serum Na+ 131 mEq/L Cl- 102 mEq/L K+ 4.7 mEq/L An x-ray of the chest shows consolidation of the left lower lobe. A Gram stain of induced sputum shows numerous neutrophils but no organisms. Which of the following is the most appropriate pharmacotherapy?"

Answer Choices:

A. Amoxicillin B. Vancomycin C. Levofloxacin D. Cotrimoxazole

Correct Answer: C. Levofloxacin

Question 202

Clinical Scenario:

A 19-year-old South Asian male presents to the family physician concerned that he is beginning to go bald. He is especially troubled because his father and grandfather "went completely bald by the age of 25," and he is willing to try anything to prevent his hair loss. The family physician prescribes a medication that prevents the conversion of testosterone to dihydrotestosterone. Which of the following enzymes is inhibited by this medication?

Answer Choices:

A. Desmolase B. Aromatase C. 5-alpha-reductase D. Cyclooxygenase 2

Correct Answer: C. 5-alpha-reductase

Question 203

Clinical Scenario:

A 23-year-old man is admitted to the hospital for observation because of a headache, dizziness, and nausea that started earlier in the day while he was working. He moves supplies for a refrigeration company and was handling a barrel of carbon tetrachloride before the symptoms began. He was not wearing a mask. One day after admission, he develops a fever and is confused. His temperature is 38.4°C (101.1°F). Serum studies show a creatinine concentration of 2.0 mg/dL and alanine aminotransferase concentration of 96 U/L. This patient's laboratory abnormalities are most likely due to which of the following processes?

Answer Choices:

A. Glutathione depletion B. Metabolite haptenization C. Microtubule stabilization D. Lipid peroxidation

Correct Answer: D. Lipid peroxidation

Question 204

Clinical Scenario:

A medical student is reviewing dose-response curves of various experimental drugs. She is specifically interested in the different factors that cause the curve to shift in different directions. From her study, she plots the following graph (see image). She marks the blue curve for drug A, which acts optimally on a receptor. After drawing the second (green) curve, she discovers that this drug B has a lower ability to produce a reaction than the first one. She also discovers that more of the second drug B is required to produce the same response as the first one. Which of the following terms best describes the activity of drug B in comparison to drug A?

Answer Choices:

A. Lower potency B. Higher potency C. Increased affinity D. Decreased efficacy

Correct Answer: A. Lower potency

Question 205

Clinical Scenario:

A 24-year-old woman delivers a girl by normal vaginal delivery, Apgar scores are 8 and 9 at 1 and 5 minutes respectively. The newborn’s vitals are normal. On examination, the attending pediatrician finds a circular skin defect that measures 0.5 cm in diameter. The defect is hairless and extends into the dermis. The delivery was atraumatic and there were no surgical instruments in the area. The pediatric team believes this is a congenital defect. The remaining examination is normal. The mother gives past history of having constant diarrhea for 3 months about 2 years ago, weight loss of 5 kg (11 lb) in 3 months, palpitations, and sensitivity to heat. She visited a community hospital and was prescribed a medication for this problem. She did not visit the hospital for any of her routine check-ups and continued taking her medications. Which drug can predispose the newborn to this condition?

Answer Choices:

A. Propylthiouracil B. Methimazole C. Propranolol D. Levothyroxine

Correct Answer: B. Methimazole

Question 206

Clinical Scenario:

A 52-year-old woman presents to her gynecologist's office with complaints of frequent hot flashes and significant sweating episodes, which affect her sleep at night. She complains that she has to change her clothes in the middle of the night because of the sweating events. She also complains of irritability, which is affecting her relationships with her husband and daughter. She reports vaginal itchiness and pain with intercourse. Her last menstrual period was eight months ago. She was diagnosed with breast cancer 15 years ago, which was promptly detected and cured successfully via mastectomy. The patient is currently interested in therapies to help control her symptoms. Which of the following options is the most appropriate medical therapy in this patient for symptomatic relief?

Answer Choices:

A. Conjugated estrogen orally B. Low-dose vaginal estrogen C. Transdermal estradiol-17B patch D. This patient is not a candidate for hormone replacement therapy.

Correct Answer: B. Low-dose vaginal estrogen

Review Summary

Category: Mechanism/Pathophysiology **Questions Reviewed:** _____ / 206

Overall Assessment: - Correctly categorized: **_ questions - Miscategorized:** ***questions - Ambiguous:*** **__** questions

General Comments:

Reviewer Name: ____________

Date: ____________

Credentials: ____________